Doctoral Thesis in Land and Water Resources Engineering

Bridging research and practice for transparent and inclusive onshore wind farm siting

Spatial multi-criteria analysis for planning support

DEEPA MANOLAN KANDY

KTH ROYAL INSTITUTE OF TECHNOLOGY



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This thesis is dedicated to everyone who encouraged me to keep moving forward and who never gave up on me throughout this journey.

To my supervisors, family, and dear friends, this milestone is dedicated to all of you.

To everyone facing challenges and refusing to give up, may you always find hope and strength.

Finally, I dedicate this thesis to myself – for having the courage to start and the determination to finish what I started.

Abstract

Sweden's ambition to achieve a fossil-free electricity system requires a substantial expansion of wind energy, which is central to climate mitigation across Europe. While necessary to reduce greenhouse gas emissions and meet growing electricity demand, wind power development involves complex trade-offs between techno-economic feasibility and socio-ecological impacts. These challenges require spatial planning approaches that integrate stakeholder perspectives and conflicting objectives. However, current applications of Spatial Multi-Criteria Analysis (SMCA) often function as technical optimisation tools rather than participatory planning processes, limiting transparency, interpretability, and real-world applicability due to methodological complexity and underlying assumptions.

This thesis aims to improve the applicability of SMCA for onshore wind energy siting in Sweden and to analyse how SMCA is currently used and how it can be enhanced. The objectives are: 1) to review SMCA applications in wind energy planning; 2) to develop and apply the REWIND framework in Västernorrland; 3) to simplify the weighting procedure through the Ranking-Rating (RR) method and analyse SMCA practice through discussions with stakeholders in Västernorrland and Västra Götaland; and 4) to refine socio-ecological and techno-economic evaluation using the Östra Mellansverige region as a case study.

The research integrates qualitative and quantitative methods within the REWIND framework, structured into three stages: scoping, design, and evaluation. Scoping involves identifying and standardising socio-ecological and techno-economic criteria using literature and stakeholder input. The design stage combines GIS-based modelling and stakeholder-derived weights to generate constraint, suitability, and conflict maps for site identification. The evaluation stage assesses alternatives using socio-ecological and techno-economic factors, including non-dominated sorting, distance-based ranking, levelied cost of energy, and annual energy yield.

Results show that SMCA applications remain dominated by techno-economic criteria, while socio-cultural and ecological dimensions are simplified and weakly integrated. The REWIND framework demonstrates that a conflict score

improves transparency by making trade-offs explicit and enabling user-defined thresholds for acceptable conflict. The framework supports structured site identification but reveals persistent challenges related to data quality, criterion representation, weighting, and uncertainty.

The RR method reduces the complexity of weighting while improving transparency and stakeholder engagement. Evaluation results show strong regional variation in wind power potential, with no correlation between techno-economic and socio-ecological performance, indicating fundamentally different dimensions of suitability. Leader-follower analysis further reveals dominance structures in some regions and more diverse solution spaces in others, with key trade-offs linked to wetlands and outdoor recreation. Overall, combining conflict analysis, participatory weighting, and structured evaluation improves transparency and supports more flexible, context-sensitive and sustainable wind power planning.

Keywords: Wind power planning, spatial multi-criteria analysis, planning support, conflict score, RR method, REWIND framework

Sammanfattning

Sverige har ambitionen att uppnå ett fossilfritt elsystem, vilket kräver en omfattande utbyggnad av vindkraft, som är central i klimatstrategier, både i Sverige och Europa. Samtidigt som vindkraft är nödvändig för att minska växthusgasutsläppen och möta en ökande efterfrågan på el, innebär dess utveckling komplexa avvägningar mellan teknisk-ekonomisk genomförbarhet och socio-ekologiska konsekvenser. Dessa utmaningar kräver rumsliga planeringsansatser som integrerar olika intressentperspektiv och hanterar motstridiga mål. Dagens tillämpningar av rumslig multikriterieanalys (Spatial Multi-Criteria Analysis, SMCA) fungerar dock ofta som tekniska optimeringsverktyg snarare än som inkluderande planeringsprocesser, vilket begränsar transparens, tolkningsbarhet och praktisk användbarhet på grund av metodologisk komplexitet och underliggande antaganden.

Denna avhandling syftar till att förbättra användbarheten av SMCA för lokalisering av landbaserad vindkraft i Sverige, samt att analysera hur SMCA används i dag och hur det kan vidareutvecklas. Målen är: 1) att granska SMCA-tillämpningar i vindkraftsplanering; 2) att utveckla och tillämpa REWIND-ramverket i Västernorrland; 3) att förenkla viktningen genom Ranking-Rating (RR)-metoden samt analysera SMCA-praktik genom diskussioner med intressenter i Västernorrland och Västra Götaland; och 4) att vidareutveckla socio-ekologisk och teknisk-ekonomisk utvärdering med fokus på regionerna inom Östra Mellansverige.

Studien integrerar kvalitativa och kvantitativa metoder inom REWIND-ramverket, strukturerade i tre steg: scoping, design och utvärdering. Scoping omfattar identifiering och standardisering av socio-ekologiska och teknisk-ekonomiska kriterier baserat på litteratur och intressentinspel. Designsteget kombinerar GIS-baserad modellering och intressentbaserade vikter för att skapa restriktions-, lämplighets- och konfliktkartor för lokalisering av potentiella vindkraftsområden. Utvärderingssteget analyserar alternativ med hjälp av socio-ekologiska och teknisk-ekonomiska indikatorer, inklusive icke-dominerad sortering, avståndsbaserad rangordning, utjämnad energikostnad och årlig elproduktion.

Resultaten visar att SMCA-tillämpningar fortfarande domineras av teknisk-ekonomiska kriterier, medan socio-kulturella och ekologiska dimensioner ofta är förenklade och svagt integrerade. REWIND-ramverket visar att ett konfliktmått ökar transparensen genom att synliggöra avvägningar och möjliggöra användardefinierade trösklar för acceptabla konfliktnivåer. Ramverket stödjer strukturerad lokalisering av vindkraft men visar samtidigt på återkommande utmaningar kopplade till datakvalitet, kriterierepresentation, viktning och osäkerhet.

RR-metoden minskar komplexiteten i viktningen samtidigt som transparensen och intressentengagemanget förbättras. Utvärderingsresultaten visar stora regionala variationer i vindkraftspotential, utan korrelation mellan teknisk-ekonomisk och socio-ekologisk prestanda, vilket indikerar att dessa representerar fundamentalt olika dimensioner av lämplighet. Analys av ledare och följare visar tydliga dominansstrukturer i vissa regioner och mer diversifierade lösningsrum i andra, med viktiga avvägningar kopplade till våtmarker och friluftsliv. Sammantaget visar resultaten att kombinationen av konfliktanalys, intressentdeltagande i viktningarna och strukturerad utvärdering ökar transparensen och stödjer en mer flexibel, kontextanpassad och hållbar vindkraftsplanering.

Nyckelord: Vindkraftsplanering; rumslig multikriterieanalys; planeringsstöd; konfliktpoäng; RR-metoden; REWIND-ramverket

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This PhD has truly been a journey. Between research and life, there were many unexpected turns and challenges over these years. Along the way, I learned not only about research but also about life, which is largely about resilience, perseverance, and the kindness of people who choose to stand beside us when I needed them most. This page is my small attempt to thank those who encouraged me to keep moving forward.

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List of publications

Paper 1: *MANOLAN KANDY, D. & MÖRTBERG, U. 2026a. A decade of spatial multi-criteria analysis for wind farm planning in Europe (in prep.).*

As first author, I conducted the literature search, selected and reviewed the documents, carried out the thematic analysis, and wrote the main parts of the manuscript, in collaboration with the co-author.

Paper 2: *MANOLAN KANDY, D., MÖRTBERG, U., WRETLING, V., KUHLEFELT, A., BYSTRÖM, G., POLATIDIS, H., BARNEY, A. & BALFORS, B. 2024. Spatial multicriteria framework for sustainable wind-farm planning – accounting for conflicts. Renewable and Sustainable Energy Reviews, 189 A, 113856.*

As the first author, I performed the data analysis, designed and implemented the methodological framework and GIS-based tools, and was the main contributor to the manuscript, in collaboration with co-authors.

Paper 3: *MANOLAN KANDY, D., MÖRTBERG, U., WRETLING, V., KUHLEFELT, & BALFORS, B. 2026b. Wind-power planning with spatial multi-criteria analysis: Lessons from exploring weighing from a planning perspective. Energy Sustainability Society (in review).*

As first author, I performed the data analysis, developed the weighting method and the Excel-based tool, implemented the approach, and was the main contributor to the writing of the manuscript, in collaboration with co-authors.

Paper 4: *MANOLAN KANDY, D., MÖRTBERG, U. & RAMIREZ GOMEZ, C. 2026c. Socio-ecological and techno-economic evaluation in SMCA-based wind farm siting in central Sweden. Renewable and Sustainable Energy Reviews (submitted).*

As the first author, I conducted the socio-ecological evaluation, including the development of the approach, analysis of results, and interpretation and discussion of findings, and contributed substantially to the writing of the manuscript, in collaboration with co-authors.

Publications not included in the thesis

MANOLAN KANDY, D. & MÖRTBERG, U (2021). A GIS-based integrated participatory approach for wind-farm siting. Oral presentation at the 16th Conference on Sustainable Development of Energy, Water and Environment Systems (SDEWES), Dubrovnik, Croatia, 10–15 October 2021.

MANOLAN KANDY, D. & MÖRTBERG, U (2022). Linking multi-criteria approaches to spatial planning for wind energy development in two case studies in Sweden. Oral presentation at the Conference on Wind Energy and Wildlife Impacts (CWW 2022), Egmond aan Zee, the Netherlands, 4–8 April 2022.

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MÖRTBERG, U, MANOLAN KANDY, D, WRETLING, V, KUHLEFELT, A, BALFORS, B. (2023). Regionalt planeringsstöd för vindkraft, Regionalt planeringsstöd för vindkraft. Naturvårdsverket Rapport 7095.

MANOLAN KANDY, D. & MÖRTBERG, U. (2025). *Wind power planning through spatial multi-criteria analysis adapted to real-world planning: A case study from Sweden*. Oral presentation at the IALE 2025 European Landscape Ecology Congress, Bratislava, Slovakia, September 2–5, 2025.

List of abbreviations

AEY	Annual Energy Yield
AHP	Analytic Hierarchy Process
CAB	County Administrative Board
EC	European Commission
EU	European Union
GIS	Geographic Information System
IEA	International Energy Agency
LCoE	Levelized Cost of Energy
NWS	National Wind Power Strategy
REWIND	The project “Regional planning for Wind power”
RR	Ranking-Rating
SDG	Sustainable Development Goals
SEA	Swedish Energy Agency
SEPA	Swedish Environmental Protection Agency
SMCA	Spatial Multi-Criteria Analysis
STA	Swedish Transport Agency
SVK	Svenska Kraftnät
UN	United Nations

WLC	Weighted Linear Combination
ÖMS	Östra Mellansverige (a seven regions coalition in eastern central Sweden)

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1 Introduction

Energy-related carbon dioxide (CO_2) emissions reached an all-time high of approximately 37.8 gigatonnes in 2024, contributing to a continued increase in atmospheric CO_2 concentrations (IEA, 2025). Although the rate of emission growth has slowed in recent years, global emissions have not yet begun to decline. This indicates that current mitigation efforts remain insufficient to flatten, and ultimately reverse, the emissions trajectory.

The continued accumulation of greenhouse gases in the atmosphere is driving significant changes in the climate system, including rising global temperatures, sea level rise, and an increase in the frequency and intensity of extreme weather events. Events such as heatwaves, wildfires, floods, tropical storms, and hurricanes are becoming more severe and frequent, with substantial impacts on society. Furthermore, projections suggest that between 2030 and 2050, climate change is expected to cause 250,000 additional deaths per year (WHO, 2023).

The energy sector is the largest contributor to global greenhouse gas emissions, making it central to climate change mitigation strategies. Within the sector, electricity and heat production represent the dominant source of emissions, accounting for a substantial share of global totals. At the same time, global energy demand has continued to increase and has been further influenced by geopolitical tensions, resource constraints, and supply chain disruptions. Together, these developments have intensified the urgency of transitioning towards renewable energy sources. Renewable energy development is strongly linked to multiple Sustainable Development Goals (SDGs), particularly affordable and clean energy (SDG 7) and climate action (SDG 13), while also involving trade-offs with biodiversity (SDG 15) and human health and well-being (SDG 3). These interlinkages highlight the need to consider both synergies and conflicts when planning renewable energy expansion.

In response, countries worldwide are increasingly shifting towards low-carbon and renewable energy systems. The renewable energy transition in Europe has been shaped most significantly by recent policy frameworks such as the Paris Agreement and the European Green Deal, which define long-term climate neutrality goals and have strongly influenced current energy system planning (EC, 2019, UN, 2015). These have been preceded by earlier milestones such as

the EU Renewable Energy Directive (2009/28/EC), which established the first binding renewable energy targets within the EU (EU, 2009).As a result, these efforts have significantly increased the deployment of renewable energy technologies. In 2025, approximately 47.3% of electricity generated in the EU came from renewable energy sources, of which wind energy contributed 47% of the total, followed by solar (27%) and hydropower (26%) (Eurostat, 2026). This increased share of renewables is a result of the ambitious, climate-friendly energy strategies adopted by EU countries. Among these technologies, wind energy has become one of the most rapidly expanding sources, driven by technological advancements, declining costs, and strong policy support across Europe. This growth has intensified the need to identify suitable locations for wind farm development while managing associated ecological and societal impacts.

In Sweden, a similar development can be observed. Wind electricity production more than tripled between 2014 and 2024, increasing from 11.2 TWh to 40.4 TWh (SCB, 2026). This rapid expansion has been supported by national planning and policy frameworks. The Swedish Energy Agency (SEA) and the Swedish Environmental Protection Agency (SEPA) developed the National Wind Energy Strategy (NWS), which guides wind power expansion while considering resource efficiency, human health, and environmental impacts. The strategy builds on initiatives from the Environmental Objectives Council and outlines pathways for achieving national energy and climate goals (Ministry of Infrastructure, 2020, SEA and SEPA, 2021).

The NWS estimates that achieving long-term energy objectives will require approximately 100 TWh of wind energy generation, of which around 80 TWh is expected to come from onshore wind power. This highlights the central role of wind energy in Sweden's future electricity system. More recently, national energy policy has been further shaped by political agreements such as the Tidö Agreement. Formulated by the governing parties (Moderates, Christian Democrats, Liberals, and Sweden Democrats) in October 2022, the agreement sets the political direction for the 2022–2026 term. It emphasises increased electricity production, a revised definition of energy as "fossil-free" rather than strictly "renewable," and anticipates a substantial rise in electricity demand, projected to reach at least 300 TWh by 2045. It also prioritises measures to accelerate the expansion of electricity infrastructure, including both generation capacity and charging infrastructure (EC, 2023).

Despite the rapid expansion of wind energy, its further development is increasingly constrained by regulatory, technical, and societal barriers. Wind power projects typically involve lengthy, complex permitting processes that often span several years and involve multiple authorities, including municipalities, county administrative boards, and the Swedish Armed Forces.

In particular, the municipal veto and national defence interests represent major sources of uncertainty and project rejection, as decisions are not always transparent and are generally not subject to appeal (Wretling et al., 2022). Approximately 66% of projects were rejected between 2020 and 2023 due to the municipal veto, rising to 75% in the first half of 2024. Notably, a substantial share of these rejections occurs at late stages of the planning process, after formal application submission, indicating high levels of uncertainty and investment risks. Furthermore, the Swedish Armed Forces have become increasingly restrictive in their assessments of wind power projects in recent years, contributing to a growing number of rejections (Westander, 2025). In addition, grid connection limitations and insufficient transmission capacity create further bottlenecks, as network expansion has not kept pace with the planned generation growth.

Together, these challenges illustrate that wind power planning is not simply a question of resource availability and energy targets, but a complex decision-making process that involves regulatory uncertainty (municipal veto, defence), spatial constraints, and conflicting socio-ecological and economic objectives. In such contexts, structured planning approaches become essential. Spatial Multi-Criteria Analysis (SMCA) is a methodological approach within Planning Support Systems (PSS) that integrates Geographic Information Systems (GIS) with multi-criteria decision analysis to evaluate multiple, and often competing, criteria in a spatial context with multiple stakeholders. It has been widely applied in the scientific literature for siting studies, such as wind farm planning, where diverse and sometimes conflicting objectives must be considered simultaneously(Malczewski and Rinner, 2015, Rediske et al., 2021). So in theory, SMCA is well-suited to meet this challenge.

Despite significant advances and the growing popularity of PSS in academia, the so-called “implementation gap” remains, indicating that these tools are still not widely or effectively integrated into planning practice (Geertman and Stillwell, 2020). Several challenges limit their practical usefulness in participatory spatial decision-making contexts. When SMCA shifts from a stakeholder-oriented planning process to a technical optimisation exercise, the process becomes opaque, difficult to interpret, and hard to reproduce due to complex methodological choices and assumptions. This is particularly problematic given that wind farm siting decisions affect multiple stakeholders across large spatial scales and have long-term implications for future generations. Core methodological components, particularly weighting and aggregation, can further obscure underlying trade-offs and provide limited support for participants in navigating conflicting objectives. In addition, the criteria included in many applications are often incomplete, oversimplified, and poorly adapted to the local context. Against this background, this thesis takes a more holistic approach by critically examining SMCA for wind energy planning,

integrating methodological, usability, and context-specific perspectives through a set of case studies.

1.1 Aim and objectives

The overall aim of this thesis is to improve the applicability of SMCA as a planning support tool for onshore wind energy siting in Sweden by developing, applying, and evaluating an enhanced SMCA framework across multiple regional case studies and stakeholder settings, while addressing methodological limitations, uncertainty, and real-world planning constraints.

The specific objectives are:

1. To critically review existing SMCA studies in wind energy planning, focusing on their suitability for real-world planning contexts. Examined themes are criteria selection and standardisation, spatial scale effects, stakeholder involvement, weighting, decision hierarchy, sensitivity analysis, and the practical uptake of SMCA outputs in planning practice (Paper 1)
2. To develop and apply the REWIND framework for onshore wind energy planning in Sweden, integrating structured scoping, design, and evaluation stages, and introducing conflict mapping to better capture planning conflicts (Paper 2)
3. To develop and test the user-friendly Ranking-Rating method for weighting of factors and analyse how participatory weighting processes influence SMCA outcomes and usability in multi-stakeholder contexts (Paper 3)
4. To refine the evaluation stage of the REWIND framework by separating techno-economic and socio-ecological assessments and reducing reliance on aggregated suitability indices (Paper 4)

1.2 Structure of the thesis

This thesis is arranged in two parts: the cover essay and the appended papers. In the cover essay, the “Introduction” (this chapter) presents the research area, followed by the aims and research questions. Chapter 2, “Background and research context”, presents concepts that underpin the work in this thesis. Chapter 3, “Methodology,” outlines the research design and methodology employed in the thesis. In Chapter 4, “Results” provides the results and discussion in relation to the literature, with the papers detailed. Chapter 5 is the final “Discussion”. Conclusions are provided in Chapter 6, and future research in Chapter 7.

2 Background

2.1 Planning support systems (PSS)

Decision-making in spatial planning becomes particularly complex when multiple, and often conflicting, criteria must be considered simultaneously. These challenges are especially evident in sustainability-related planning problems, where environmental, social, and economic objectives must be balanced under uncertainty and spatial constraints. To support such complex decision-making, PSS have been developed. These are computer-based tools that assist planners in performing their day-to-day professional tasks more effectively (Pelzer, 2017, Krishnan et al., 2021). They add value to planning processes by integrating a range of components, including spreadsheets, websites, GIS, visualisation tools, and modelling systems. Initially, PSS emerged in response to strong criticism of earlier top-down, comprehensive, and "black-box" models used to optimise spatial development (Geertman, 2006, Geertman and Stillwell, 2020)

Over time, PSS have evolved alongside broader changes in planning theory and governance. Up to the 1970s and 1980s, spatial planning was largely a rational, expert-driven activity in which professional planners relied on scientific knowledge to design long-term "blueprints" for the spatial organisation of society. This approach was underpinned by an optimistic belief that society and space could be shaped in a predictable and controllable manner. However, from the 1980s onwards, this perspective gradually shifted amid increasing societal complexity, including democratisation, economic restructuring, and uncertainty in development pathways. This led to a transition from blueprint planning to more process-oriented and collaborative planning approaches, often referred to as the communicative turn. In this context, planning became understood as a participatory process involving a wide range of actors, including public authorities, private stakeholders, non-governmental organisations, and civil society, while knowledge itself came to be viewed as socially constructed rather than purely objective (Geertman and Stillwell, 2020).

2.2 Swedish wind power planning:

Wind power planning in Sweden stands out as one of the most decentralised planning systems in Europe, with responsibilities distributed across national, regional, and local levels (Wretling et al., 2022). Wind farm projects are typically initiated by private developers, who identify suitable sites through spatial analyses of wind resources, environmental constraints, and grid accessibility. Developers are responsible for preparing and submitting permit applications for large wind farms, including an Environmental Impact Assessment (EIA), in accordance with the Swedish Environmental Code. Large wind farms are defined as installations consisting of two or more turbines exceeding 150 metres in height, or seven or more turbines exceeding 120 metres, including rotor blades. Such projects require municipal approval, an environmental permit, and a completed EIA (Abboud, 2021).

At the municipal level, a key planning instrument is the comprehensive plan (översiktsplan), which outlines long-term strategies for land and water use. Municipalities also have the option to adopt thematic amendments to the comprehensive plan in order to address specific issues not fully covered in the original plan (Wretling et al., 2022). Although comprehensive plans are not legally binding, they play an important role in indicating suitable areas for wind energy development and in guiding both municipal decision-making and developer site selection (Abboud, 2021).

A formal requirement in the Swedish system is municipal approval of wind power projects, which is commonly referred to as a municipal veto. In practice, this means that municipalities can prevent a wind power project from being authorised even if it meets environmental requirements. There is no legal obligation for municipalities to justify their decision, nor any fixed requirement regarding when during the permitting process the decision must be made (Wretling et al., 2022). This gives municipalities a decisive role in determining whether projects can proceed.

At the regional level, County Administrative Boards (CAB) are responsible for coordinating environmental assessment processes and ensuring that national interests are considered. Within these boards, the Environmental Assessment Delegation (Miljöprövningsdelegationen) serves as the main permitting authority for onshore wind power projects. It assesses applications under the Environmental Code, with a particular focus on environmental impacts on nature conservation, cultural heritage, and other protected interests. The assessment is based on the Environmental Impact Assessment and supporting documentation submitted by the developer (Ahonen-Jonnarth et al., 2023).

In addition to environmental considerations, Areas of National Interests (ANI, riksintressen), such as infrastructure, environmental protection, and defence, are integrated into the permitting process. The Swedish Armed Forces represent

a key stakeholder and may object to wind power projects that interfere with radar systems, flight routes, or other military infrastructure (Odell et al., 2022). Overall, this permitting process is multi-layered, time-consuming, and characterised by a high degree of uncertainty.

2.3 Spatial Multi-Criteria Analysis (SMCA)

Within planning support environments such as PSS, structured analytical methods are used to evaluate complex decision problems involving multiple criteria. One of the most widely applied approaches is Multi-Criteria Decision Analysis (MCA), which has long historical roots. An early example is moral algebra by Benjamin Franklin in 1772. When his friends was facing a difficult choice, Franklin suggested listing pros and cons in two columns, assigning relative importance to each over several days, and systematically eliminating items of comparable weight (Maas, 2016). This method represents a simple yet systematic way of balancing competing considerations to reach a rational decision.

Over time, more formalised approaches have been developed. When Multi-Criteria Analysis (MCA) is applied in a spatial context, it is called Spatial Multi-Criteria Analysis (SMCA). The spatial dimension is critical, as the suitability or value of a decision may vary depending on location, providing the key motivation for such analyses.

MCA can be divided into two main categories: Multi-Attribute Decision Analysis (MADA) and Multi-Objective Decision Analysis (MODA). MADA typically addresses choice problems involving a finite set of predefined alternatives, where each alternative is evaluated and ranked based on known attributes such as location, size, or other characteristics. In contrast, MODA focuses on search and optimisation problems, where the goal is to identify optimal solutions without necessarily having predefined alternatives. These solutions can be a combination of dominated and non-dominated solutions and represent trade-offs where improvement in one objective would lead to deterioration in another (Malczewski, 2006, Malczewski and Rinner, 2015). In essence, MODA relies on objectives as the primary drivers of decision-making, whereas MADA begins with a set of alternatives evaluated against a given set of criteria.

A fundamental challenge in SMCA is the identification of suitable alternatives, commonly referred to as the siting problem. Applications include the placement of landfills, solar farms, wind farms and biomass facilities (Asefi et al., 2020, Giamalaki and Tsoutsos, 2019, Perpiña et al., 2013). In spatial terms, a “site” can be understood as a defined area characterised by both spatial and non-spatial attributes. In raster-based analyses, sites are represented as aggregations of pixels or cells. The objective is to identify clusters of cells that satisfy predefined criteria and constraints (Cova and Church, 2000).

Such siting problems often resemble MODA, as predefined alternatives may not exist and the characteristics of potential sites are initially unknown. However, they can also be interpreted as MADA problems for several reasons. First, raster-based analyses inherently deal with a finite number of discrete units (pixels), each of which can be treated as an alternative. Second, the process is outcome-oriented, aiming to identify the most suitable locations among available options. Third, most siting studies employ SMCA methodologies rather than computationally intensive optimisation techniques typically associated with MODA (Malczewski and Rinner, 2015). In this context, the search for potential sites is conceptualised as a *design problem*, while the comparison of selected alternatives is treated as an *evaluation problem* (Manolan Kandy, 2023).

Within MCA and SMCA, criteria represent the key factors influencing decision-making. In this case, the focus is on the sustainable siting of wind farms. These factors may include, for example, wind resource availability (for optimal turbine performance), noise impacts on residential areas, and the presence of sensitive bird species that may be affected by turbine blades and flight paths. In spatial applications, such criteria are operationalised as geographic datasets (e.g., wind speed distribution maps, distance to settlements, or habitat suitability), typically represented as maps. These spatial representations form the basis for subsequent analysis and must be transformed into a comparable scale to support aggregation.

To enable comparison and aggregation, criteria must be transformed to a common scale through a process known as standardisation or value scaling. This step converts raw criterion values into dimensionless suitability scores that reflect the decision-maker's preferences. In this context, value functions are used to map criterion values to a standardised scale, typically ranging from 0 (least suitable) to 1 (most suitable), thereby providing a mathematical representation of desirability (Keeney, 1980).

Standardisation can be broadly categorised into crisp and fuzzy approaches. Crisp standardisation assigns Boolean values (e.g. suitable/unsuitable) based on predefined thresholds and is typically used to represent constraints. In contrast, fuzzy approaches based on the theory of fuzzy sets (Zadeh, 1999) allow for gradual transitions between suitability levels by expressing the degree of membership of a location to a suitability class. This enables a more flexible representation of uncertainty and vagueness in decision criteria and is widely applied in GIS-based multi-criteria analysis. The shape of the value function reflects how preferences change across the range of a criterion. While linear transformations are commonly used in GIS-based SMCA for their simplicity, non-linear forms can also capture different preference structures, such as risk-averse or risk-seeking behaviour. In practice, value functions are often implemented as piecewise linear approximations based on expert judgement.

Among the various aggregation methods used in SMCA, the Weighted Linear Combination (WLC), also referred to as simple additive weighting, weighted overlay, or weighted aggregation, is one of the most widely applied approaches (Malczewski and Rinner, 2015). WLC is a map algebra–based aggregation procedure in which standardised factor maps are multiplied by weights (often derived from stakeholders) and summed to produce an overall suitability score.

2.3.1 Other evaluation methods:

There are also many other methods that can be applied when there is a limited number of predefined alternatives to choose from. Some of the most prevalent methods are Preference Ranking Organisation Method for Enrichment Evaluation (PROMETHEE) (Brans and Vincke, 1985), Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) (Hwang and Yoon, 1981)and ELimination Et Choix TRaduisant la REalité (ELECTRE) (Roy et al., 1986).

PROMETHEE belongs to the outranking school of MCA methods (Behzadian et al., 2010), with the version PROMETHEE II being widely used in the evaluation stage of wind farm siting. The core of this method is the pairwise comparison of each alternative based on the different criteria and the ranking is based on net flow (Mateo, 2012).

The method TOPSIS belongs to the Ideal Point School of methods (Malczewski and Rinner, 2015) and works on the principle that for a good solution, the Euclidean distances from the ideal solution should be the shortest, and the distance from the worst solution should be the longest. The method assumes that the criteria in the decision matrix are either monotonically increasing or decreasing (Hwang, 1981). This method has been applied in combination with AHP, PROMETHEE or other methods to determine the ranking of sites in wind projects (Gigović et al., 2017, Sánchez-Lozano et al., 2014).

3 Methodology

The research was conducted within the context of two projects: **REWIND – regional planning for wind power**, focusing on the counties of Västernorrland and Västra Götaland, and **Wind-ÖMS – potential for wind power in the Östra Mellansverige regions**. In addition, the **Wind-GIS** project conducted within Västernorrland County was part of a previous master's thesis (Manolan Kandy, 2018) that provided an initial starting point for this thesis regarding the SMCA application and stakeholder weighting. These projects provided complementary methodological foundations for this thesis.

The **REWIND** project primarily focused on identifying suitable sites for wind farm development, considering economic, social, and ecological aspects. In this thesis, these are termed the scoping and design stages of the SMCA, comprising the identification and standardisation of criteria, the development of weighting approaches, and the assessment of suitability and conflict scores. This stage was also informed by stakeholder engagement through focus group discussions, which aimed to incorporate real-world planning challenges into the SMCA framework.

In contrast, the **Wind-ÖMS** project placed greater emphasis on evaluating selected alternatives, corresponding to the *evaluation stage* of the SMCA. This included structuring decision alternatives to improve transparency, as well as techno-economic evaluation of these alternatives. The focus was on addressing practical planning challenges within the spatial planning for wind energy. Similar to the REWIND project, stakeholder discussions played an important role in informing the analysis and ensuring the relevance of the methodological approach.

The methodological components applied across the four papers included both qualitative and quantitative methods: literature review, study area analysis, focus group interviews and discussions, and the application of the SMCA framework using GIS-based modelling. Each paper addressed these components differently depending on the research focus, as summarised in Table 1.

Table 1: Overview of methods applied in the papers.

	Paper 1	Paper 2	Paper 3	Paper 4
Literature review	√	√	√	√
Focus group interviews and discussions		√	√	√
Case study		√	√	√
SMCA framework	General SMCA	Design focus	Design focus	Evaluation focus

3.1 Literature review

A literature review helps develop research by building on existing knowledge. In the field of SMCA on wind power planning (Wind-SMCA), knowledge production has increased significantly in recent years (Malczewski and Rinner, 2015). However, due to its interdisciplinary nature—spanning spatial planning, social sciences, environmental sciences, decision science, and energy systems—this body of knowledge remains fragmented. This fragmentation creates a need for a systematic literature review to identify relevant approaches and highlight existing research gaps. Literature reviews are therefore used as a methodological tool across all four papers in this thesis.

The research began with an initial literature review to identify relevant criteria influencing wind power planning using SMCA and to explore existing methodological approaches. This early mapping, although not fully systematic, helped structure the subsequent research. The literature review informed the design stage of Papers 2 and 3, during which criteria identification, factor standardisations, and methodological approaches were developed.

Paper 1 was a systematic review of onshore wind SMCA studies within Europe from 2014 to 2024. Peer-reviewed articles were retrieved from Web of Science Core Collection and Scopus using a structured search query with the corresponding database help. The query used in Web of Science Core Collection was: (TS = ("wind power" OR wind power OR "wind energy" OR "wind farm*") AND TS = ("multi criteria" or multicriteria or MCDA or MCA or MCE or SMCA or SMCE or "spatial multi criteria" or GIS) AND TS = (Site or location or locate or spatial)). TS is the field tag used in the query to search for topic terms in the title, abstract, author keywords, and keywords plus. The Boolean operators "OR" and "AND" were used to combine search terms. The review focused on themes grouped into two overarching categories: analytical methods (criteria, weighting methods, sensitivity analysis, decision hierarchy) and planning context (year and country, spatial scale, stakeholder involvement, real-world planning uptake).

Finally, Paper 4 applied these methodological insights from literature reviews to evaluate planning alternatives in central Sweden, integrating socio-ecological and techno-economic factors. Literature from techno-economic studies, guided by the Energy Systems Department at KTH, was also integrated to support the methodology and contextualisation in this paper.

3.2 Focus group interviews

The main features of focus group interviews and discussions are the collection of data and gathering of knowledge through structured group interaction on a pre-defined research theme, facilitated by a moderator. The shared meanings expressed in the group provide insights into the knowledge, perceptions and attitudes of the selected participants (van Bezouw et al., 2019). This is important because SMCA-based planning is not purely technical but involves several stakeholders, each with their own value judgements and planning experience. Focus group interviews and discussions were used in Papers 2, 3, and 4.

- Focus group interviews in the Wind-GIS project – Paper 2 (weights)
- Focus group interviews in the REWIND project – Paper 2 and Paper 3
- Focus group interviews in the Wind-ÖMS project – Paper 4

Together, these discussions formed a sequential, iterative knowledge base in which insights from the Wind-GIS and REWIND projects informed the implementation of the Wind-ÖMS project.

Note: The Wind-GIS project was part of my master's thesis (Manolan Kandy, 2018), and while it contributed to weighting for Paper 2, it is only briefly described here; the focus of this thesis is on the REWIND and Wind-ÖMS focus group discussions.

3.2.1 Participants in the focus group interviews

The participants in the focus group interviews represented key institutions and areas of expertise relevant to wind power planning. Participants were selected through existing project networks associated with the REWIND and Wind-ÖMS projects. In addition, a snowball sampling approach was used, in which initial participants recommended additional relevant stakeholders to ensure representation across sectors and governance levels.

Many participants were present across sessions and had prior experience working with SMCA in planning. This ensured continuity and more in-depth engagement in the discussions. Overall, participants brought expertise in data, energy systems, spatial planning, environmental challenges and assessment, and policy, enabling integration of both practical planning experience and strategic perspectives.

In the REWIND project, participants were primarily from the County Administrative Boards, including the environmental permitting bodies, and from the Regions. They were planners, GIS coordinators, energy and climate strategists, and environmental officials. The group was complemented by KTH researchers who held presentations and moderated discussions. The focus group discussions were conducted online in 2021 and consisted of three sessions for each of two regional case studies: Västernorrland and Västra Götaland. Each session lasted approximately three hours, including a short break, and involved around 8–10 participants.

In the Wind-ÖMS project, the composition of participants was similar but slightly broader, including representatives from the Swedish Energy Agency, Region Stockholm, County Administrative Boards and the company Vattenfall. Four focus group sessions were conducted in 2024 and 2025, each lasting approximately 4 hours. The sessions were held at KTH Royal Institute of Technology, with the possibility for hybrid participation. In total, around 10 participants were involved, with attendance varying between sessions.

3.2.2 Implementation in the projects

Focus group interviews - REWIND project (Papers 2 and Paper 3)

Focus group interviews were central methodological components in the development and application of the REWIND framework, reported in Paper 2 and Paper 3. In 2021, three focus group sessions were conducted for each of the two case study regions. Each session lasted approximately three hours and was held in a digital format. A total of 16 stakeholders participated, representing County Administrative Boards, Environmental Appeal Delegations, and Region Västra Götaland, as well as four KTH researchers acting as facilitators and note-takers.

The discussions served two main purposes:

i. to gather stakeholder knowledge on region-specific factors, preferences, and concerns within the SMCA process, and
ii. to evaluate the applicability of the RR weighting method as a planning support tool in real-world wind power planning contexts.

Prior to each session, participants received preparatory materials, including agendas, factor catalogues, planning tree structures, and methodological descriptions. Each session began with a presentation followed by structured questions and discussions.

Session 1 (scoping stage): Focused on identifying relevant factors, refining the factor catalogue, discussing data availability, and exploring clustering options within the planning tree. Questions:

- Which criteria should be used for wind power planning?
- How should these be organised?
- What data should we use to obtain spatial indicators (maps) for the criteria?
- How should data be treated to best reflect the criteria?
- Should we conduct the analysis at a smaller scale, or include all seven regions in ÖMS?
- What size of wind farms are we targeting?

Session 2 (scoping stage): Centred on the weighting exercise using the RR method. Stakeholders worked in smaller breakout groups using an Excel-based tool, each adopting a different professional perspective to assign weights. Discussions addressed both the weighting process and broader planning considerations.

- Which factor do you consider most important based on your professional experience?
- How much more important is it compared to the others?
- How did you find the weighting exercise?
- Do the final weights reflect your overall preferences?
- Should we develop multiple scenarios?
- Should techno-economic and socio-ecological aspects be assessed separately?

Session 3 (design and evaluation stages): Focused on reviewing results, including weight distributions, suitability maps, and proposed planning alternatives. Discussions explored the applicability of both the RR method and the REWIND framework in practical planning contexts, including their role in stakeholder dialogue processes.

The outcomes from these sessions directly informed the generation of suitability maps.

- What are your views on the final suitable sites?
- How can the SMCA framework be improved further?
- What planning challenges did we encounter during this process?

Focus group interviews - Wind-GIS project (Paper 2)

Focus group interviews and discussions were also conducted as part of the Wind-GIS project, in which I participated during my master's thesis in 2018. These sessions took place on 12 June and 26 October 2018 at the County

Administrative Board of Västernorrland in Härnösand and involved 14 participants from the Swedish Energy Agency, CAB Västernorrland, and KTH Royal Institute of Technology. The discussions focused on structuring the planning tree and conducting a weighting exercise. An online AHP tool, the AHP Priority Calculator (Goepel, 2018a, Goepel, 2018b), was used to derive factor weights. The resulting aggregated weight profile, calculated as the arithmetic mean of individual weights, was subsequently applied in the design stage of the REWIND framework and informed the analysis presented in Paper 2.

Focus group interviews - Wind-ÖMS project (Paper 4)

In Paper 4, focus group interviews were conducted within the Wind-ÖMS project, with a primary emphasis on the evaluation stage of the planning process, particularly concerning socio-ecological and techno-economic factors. Four in-person sessions were held at KTH during 2024 and 2025, each lasting approximately four hours. A total of 10 stakeholders participated, representing the Swedish Energy Agency, Region Stockholm, County Administrative Boards, in addition to the KTH researchers.

The sessions were structured as follows:

- Sessions 1–2 (scoping stage): Focused on data availability, treatment of factors, and the identification of socio-ecological and techno-economic criteria. Discussions also included refinement of the factor catalogue and initial weighting considerations using the RR method.
- Session 3 (design and evaluation stages): Addressed the development of planning scenarios and interpretation of weighting results, including exploration of alternative scenario configurations.
- Session 4 (evaluation stage): Focused on discussion of results, evaluation approaches, and key insights regarding the applicability of the framework and its outcomes.

The planning alternatives evaluated in Paper 4 were derived from earlier focus group discussions (as in Paper 2), ensuring methodological consistency across the research. The questions were the same as described in the REWIND project (Paper 2). An overview of the focus group sessions is provided in Table 2.

Table 2: Overview of focus group sessions in the REWIND and Wind-ÖMS projects.

Stage	Focus	Sessions	Method	Output	Paper contributions
Scoping	Identifying, structuring and standardising socio-ecological and techno-economic criteria	REWIND sessions 1,2,3 Wind-ÖMS sessions 1 and 2	Review of factor catalogues, exchanges of spatial data, development of factor building tool, group discussions	Updated factor catalogue	Paper 2,3 and 4
Weighting	Prioritise factors	Wind-GIS, REWIND session 2 and Wind-ÖMS session 3	AHP online tool and RR method in Excel	Weights for factors	Paper 2,3 and 4
Scenarios	Develop planning alternatives	Wind-ÖMS session 3	Scenario discussions (Baseline scenario, distributive justice focus with small farms, and energy focus with big farms)	Planning scenarios	Paper 4
Evaluation of alternatives	Evaluate the selected alternatives	Wind-ÖMS session 4	Result interpretation and comparison of results with theirs	Insights about structuring the alternatives	Paper 4

3.3 Case studies

According to Yin (2009), " a case study is an empirical inquiry that investigates a contemporary phenomenon within its real-life context, especially when the boundaries between phenomenon and context are not clearly evident" (Yin, 2009). This thesis adopts a case study approach, in which selected study areas provide context for the application of the SMCA framework. The outcomes of SMCA applications are inherently context-dependent, shaped by geographical conditions, stakeholder perspectives, and the practical challenges identified

within each study area. The framework was applied across the study areas, in all the papers except Paper 1. Study areas were from the SE2 and SE3 price zones of Sweden, see Figure 1.

Case study 1 focused on Västernorrland County (SE1) (Paper 2), while Case study 2 included both Västernorrland (SE2) and Västra Götaland (SE3) (Paper 3). Case study 3 covered seven counties (regions) in east-central Sweden (SE2 and SE3), collectively referred to as the ÖMS Regions (Paper 4). These case studies differ substantially in terms of electricity production, population density, geographical conditions, and the number of wind farms. They also reflect varying stakeholder perspectives on wind power planning.



Figur 1. Study areas applied to Paper 2 (P2), Paper 3 (P3) and Paper 4 (P4).

3.3.1 Västernorrland (Paper 2 and Paper 3)

Västernorrland county, located in northern Sweden along the coast of the Baltic Sea in the Price zone SE2 (see Figure 1), was used as the case study in Papers 2 and 3. The county comprises seven municipalities and covers a total land area of approximately 23,084 km². The landscape is diverse, featuring a long coastal zone in the east, forested and mountainous terrain in the west, and numerous lakes and rivers throughout the region. Land cover is dominated by forests (approximately 81%), followed by other open land (8%), water bodies (7%), agricultural land (2%), and urban and infrastructure areas (2%) (SEPA, 2018a). The county is sparsely populated, reflecting relatively low land-use intensity.

The electricity production in Västernorrland is primarily based on hydropower (85%), followed by wind power (7%), industrial backpressure (7%), and a minor contribution from solar energy (1%) (SCB, 2023a). Wind power generation in this county has increased significantly in recent years, reaching approximately 8.2 TWh in 2023 (SCB, 2022). The combination of low population density, high wind energy generation, and substantial land availability make Västernorrland a relevant case for exploring wind power planning in seemingly less-constrained environments.

3.3.2 Västra Götaland (Paper 3)

Västra Götaland county, located in south-western Sweden along the coast of the North Sea in Price zone SE3, was used as a case study in Paper 3. Although similar in size to Västernorrland, it is much more densely populated and comprises 49 municipalities, with notable differences in land use (Lantmäteriet, 2020) and energy demand (SCB, 2023a). This county has the second-highest population in the country after Stockholm County and has a population density of 74.3/km² (SCB, 2023b). The diverse landscape with varied terrain consists of forest, agricultural areas, mountains, plateaus and archipelago. It consists of about 61% forest and wetlands, 18% arable land, 9% other open land, 8% water, and 4% urban land and infrastructure (SEPA, 2018a). Different sources of electricity generation like wind power (47%), hydropower (41%), combined heat and power and industrial backpressure (13%) and other thermal power (0.01%) were used in 2020 in this county (SCB, 2022). The combination of high energy demand, earlier wind development, and intensive land use makes Västra Götaland an interesting case, providing a contrast to the sparsely populated Västernorrland County.

3.3.3 Central Sweden – the ÖMS Regions (Paper 4)

In Paper 4, the study area encompassed seven regions in central Sweden: Gävleborg, Uppsala, Västmanland, Stockholm, Södermanland, Örebro, and Östergötland. The regions have the same borders as the counties, but form separate governance bodies. The 7 regions in this case study collaborate under the umbrella of the ÖMS Regions, a coalition with which we engaged due to their interest in renewable energy. The total land area is approximately 70,263 km². Gävleborg Region has the largest total area, a high proportion of forest, low shares of agricultural and urban land, and low population density, reflecting characteristics typical of northern Sweden. In contrast, the Stockholm Region, which includes the national capital, is characterised by a high share of urban land and the highest total population and population density among the seven regions. The regions Uppsala, Västmanland, Södermanland, and Östergötland have relatively extensive agricultural areas.

Most of the regions are located within the SE3 electricity price zone (see Figure 1), which is characterised by high electricity demand and limited generation capacity. Gävleborg spans both the SE2 and SE3 price zones, with most of its area in SE2, where electricity generation is higher, and demand and prices are lower. Wind farm distribution across the study area reflects these differences: northern counties benefit from less intensive land use, leading to larger wind farms with taller turbines, whereas southern counties face stricter planning constraints, higher land-use pressures, and more complex socio-ecological considerations (Vindbrukskollen, 2026). To contextualise regional differences in wind power potential, Table 3 presents a county-level comparison of NWS-based estimates alongside wind electricity production, total electricity generation, electricity demand, and population density in 2023.

Table 3: County-level comparison of wind power potential and electricity data (2023)

Region	NWS area estimates (km^2)[1]	NWS energy distribution (TWh)[1]	Wind electricity production 2023 (TWh)[2]	Electricity production 2023 (TWh)[2]	Electricity usage 2023 (TWh)[2]	Population density 2023 (Total population/km^2)[3]
Västernorrland (SE2)	338	7.5	8.2	19.7	8.31	11.2
Gävleborg (SE2 and SE3)	338	7.5	3.08	5.5	4.9	15.8
Uppsala (SE3)	113	2.5	NoData	25.9	3.3	49.4
Västmanland (SE3)	90	2	NoData	0.8	2.8	54.9
Stockholm (SE3)	90	2	NoData	NoData	NoData	376.9
Södermanland (SE3)	90	2	0.72	0.5	3.3	49.8
Örebro (SE3)	113	2.5	NoData	NoData	3.8	36.2
Östergotland (SE3)	113	2.5	NoData	2	6	44.7
Västra Götaland (SE3)	338	7.5	3.16	6.5	18.14	74.3

[1]SEA and SEPA (2021)
[2]SCB (2023a)
[3]SCB (2023b)

3.3.4 Spatial distribution of wind farms in the study areas

Existing wind farms

Counties in Price zone SE2, such as Västernorrland and Gävleborg, have a high number of existing wind turbines: 836 and 315, respectively. These counties are relatively sparsely populated and have lower land use intensity. In contrast, Västra Götaland, located in SE3, is an interesting case, with 658 turbines despite its high population density and more intensive land use. Within the central Sweden (ÖMS) study area, Östergötland and Örebro have a relatively higher number of turbines than other counties in the area. At the same time, there is a large variation across the ÖMS Regions, ranging from as few as 11 turbines in Uppsala to 162 in Östergötland (Vindbrukskollen, 2026). This highlights a significant spatial difference in wind power development, even among neighbouring counties with similar population densities and within the same electricity price zone.

When comparing wind farms in Västernorrland and Västra Götaland, there are only 45 wind farms in Västernorrland, whereas Västra Götaland has 372 wind farms. Simultaneously, wind farms in Västra Götaland are smaller and more dispersed, while in Västernorrland they are larger and more consolidated, reflecting differences in population density and land-use patterns between the two counties. Also, the number of turbines exceeding 200 m in total height is 279 in Västernorrland, compared to only 7 in Västra Götaland, further highlighting differences in turbine heights (Vindbrukskollen, 2026).

Rejected and appealed wind farms

The distribution of rejected and appealed wind power projects varies considerably across the counties, as shown in Table 4 (Vindbrukskollen, 2026). Västra Götaland has the highest number of appeals (334), exceeding Västernorrland's 227, likely due to its higher population density and intensive land use. Östergötland also shows many appeals (105) despite a moderate number of existing turbines. In contrast, Stockholm and Västmanland reported very few appeals (1 and 3), reflecting limited project applications. Overall, the number of appeals appears to be influenced not only by the number of existing turbines but also by population density, land-use intensity, and regional planning constraints.

Table 4. Number of existing, appealed and rejected turbines across all the counties included in this thesis

County	Number of existing turbines	Number of appeals and rejections
Västernorrland	836	227
Gävleborg	315	134
Uppsala	11	7
Västmanland	25	3
Stockholm	24	1
Södermanland	17	24
Örebro	105	35
Östergötland	162	105
Västra Götaland	658	334

3.4 REWIND methodological framework

The SMCA framework applied in this thesis, the REWIND methodological framework (Paper 2), was developed to support a systematic approach to wind farm site selection from the initial criteria that impact wind power planning to final decision-making. The framework was designed for application at both regional and municipal levels and integrates spatial data with stakeholder preferences throughout the planning process. It has been progressively developed and applied across the included papers. An overview of the framework is presented in Figure 2, consisting of three main stages: scoping, design, and evaluation.

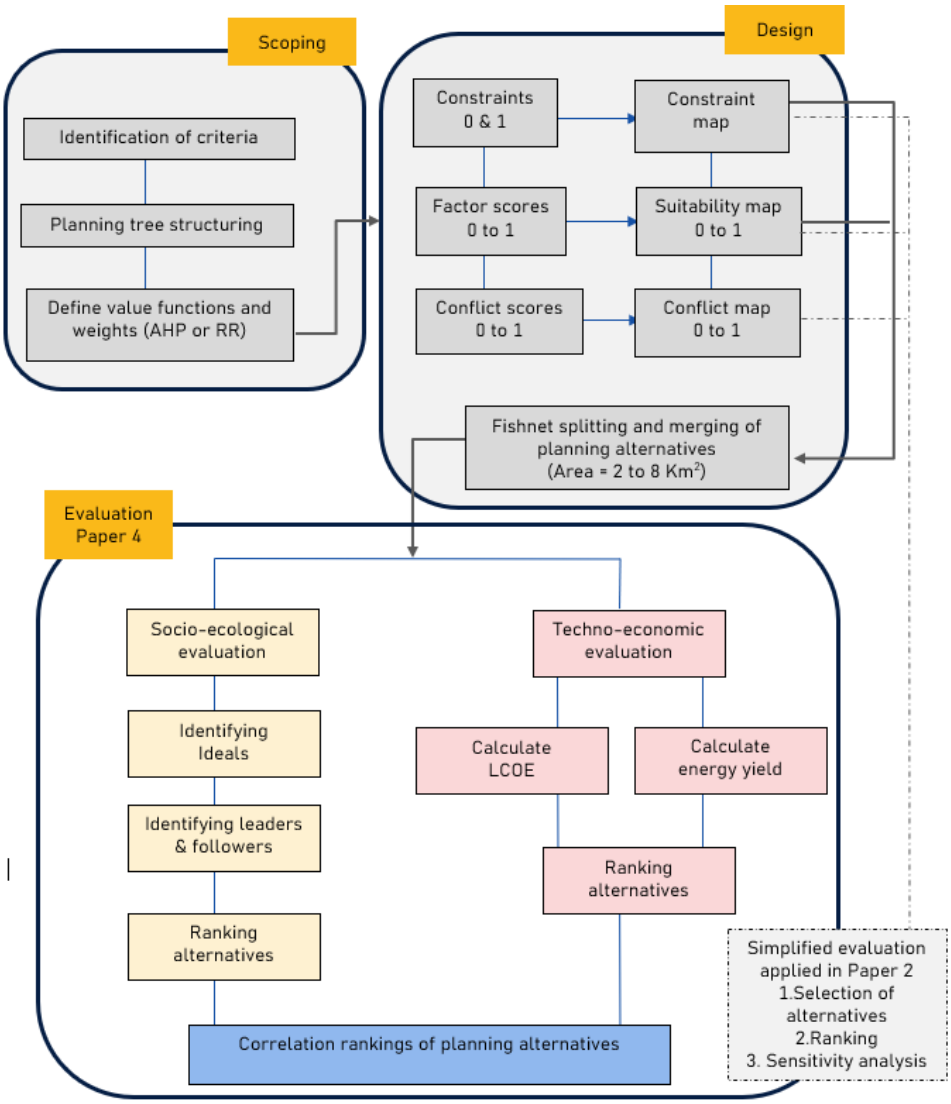


Figure 1: Overview of the REWIND methodological framework.

In the **scoping stage**, relevant criteria are identified, structured into a planning tree, and spatial indicators for the criteria are derived and standardised to a single scale for further analysis. During this stage, stakeholder preferences among the criteria are also obtained through weighting exercises.

In the **design stage**, spatial datasets and stakeholder-derived weights are combined to generate constraint, suitability and conflict maps, from which potential planning alternatives are identified.

In the **evaluation stage**, these alternatives are compared and assessed to support decision-making.

In Papers 2 and 3, the framework was primarily applied to develop and test methodological components during the scoping and design stages, including planning tree structuring, factor standardisation, weighting approaches, and the use of conflict scores. The evaluation step for Paper 2 was basic, involving ranking based on suitability and conflict scores, along with a weight-alteration sensitivity analysis, as illustrated in Figure 2.

In Paper 4, the framework was further extended, with a particular focus on the evaluation stage. In this stage, socio-ecological and techno-economic perspectives were integrated to enable a more comprehensive assessment of planning alternatives, with the aim of improving transparency around trade-offs and bridging the perspectives of planners and developers.

3.4.1 Scoping stage

General scoping stage

The general scoping stage across all case studies involved identifying, structuring, and standardising criteria, and obtaining stakeholder weights for relevant factors. Influencing criteria were derived from scientific literature, focus group discussions, relevant legislation, and grey literature (Wretling et al., 2022). Once identified, the criteria and their corresponding spatial indicators were standardised into a common suitability scale using fuzzy logic, Boolean logic, or a combination of both, depending on the context of each case study. Fuzzy logic was applied to factors whose suitability varies gradually from 0 to 1 and is not strictly constrained by law or other binding requirements, whereas constraints were mapped to either 0 or 1 to represent areas where development is strictly prohibited by law or other binding requirements. Decisions regarding standardisation were informed by policy, legal requirements, scientific literature, and stakeholder input, ensuring relevance for planning and decision-making.

Due to the large number of criteria, they were organised into a hierarchical structure, called a planning tree, as in Figure 3. The clustering of these factors and constraints was based on the type of impact, legislative relevance, and alignment with the Sustainable Development Goals. The planning tree provided a structured overview of the decision problem and simplified the weighting process. Although there is no standardised format for such hierarchies, the clusters applied across the case studies typically included wind power

development, socio-cultural values, nature conservation, reindeer herding, and other land use and infrastructure. Minor adjustments were made between case studies to account for regional relevance, such as the inclusion or exclusion of reindeer herding or specific bird species.

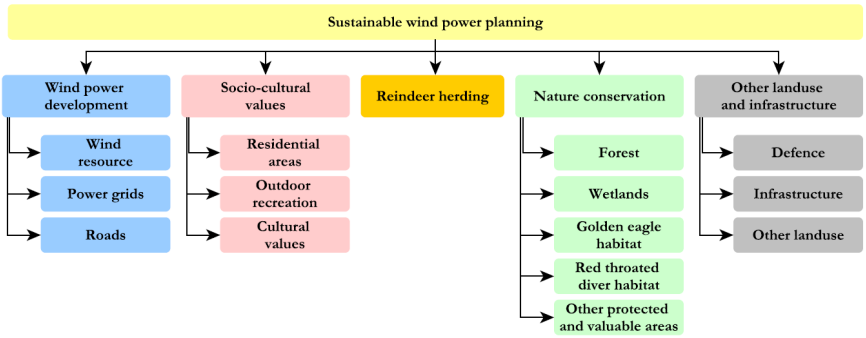


Figure 2: A planning tree with clusters of criteria applied in the case of Västernorrland.

For the raster-based spatial analyses, a 25 × 25 m cell size was used. Factor maps were generated by combining sub-factors within the same cluster, with decisions on aggregation (e.g., minimum or maximum suitability) guided by scientific literature and stakeholder recommendations. Overall, stakeholders played a central role in the scoping process.

Data and factor development

The study used a combination of spatial datasets applied in the REWIND framework. A full list of datasets, sources, and the standardisation procedure, including control points, is provided in Table 5. In total, approximately 90 spatial datasets were used, from various sources. While many datasets were common across the case studies, some were modified to reflect regional requirements, such as the inclusion of reindeer herding in Västernorrland. The standardisation approach was also adapted in response to stakeholder input. For example, in Paper 3, wind speed was standardised differently between Västernorrland (VN) and Västra Götaland (VG): areas with wind speeds below 6 m/s were assigned a suitability score of 0 in VG, whereas this threshold was not applied in VN. This difference reflected the greater emphasis on economic feasibility in VG, as well as the availability of generally stronger and more consistent wind resources along the western coast compared to the more spatially heterogeneous wind conditions in VN, which influenced how wind-speed suitability was assessed.

If we look at the factor reflecting residential areas, under the socio-cultural factor cluster, it also includes places which people use temporarily, such as community houses, hospitals, preschools, and holiday homes. Sub-factors were identified through online searches and, in some cases, data was supplemented with stakeholder input. Obtaining data for certain infrastructure, such as regional lower-voltage grids, was challenging. Consequently, only regional 130 kV grids were used in the analyses. In Paper 4, the techno-economic evaluation was conducted in greater detail during the evaluation stage, leading to differences in how these factors were standardised compared to the other case studies.

Table 5. Spatial datasets and standardisation approaches applied across Papers 2 (P2), 3 (P3), and 4 (P4). For Paper 3, distinctions are made between case studies in Västernorrland (P3-VN) and Västra Götaland (P3-VG)

Wind power development	
Wind resource: Wind speed at a height of 140 m (P4), wind speed at 120 m (P2), Wind speed at 150 m (P3) (Bergström and Söderberg, 2012)	Below 3 m/s score 0.01, linear increase to 1 for 7 m/s and above (P4) Below 5 m/s score 0, linear increase to 1 for 7 m/s and above (P2) Below 5 m/s score 0, 5-6 m/s score 0.01, linear increase to 1 for 12 m/s and above (P3-VN) Below 6 m/s score 0, linear increase to 1 for 12 m/s and above (P3-VG)
Electricity grid: Regional electricity grid lines (P2, P4) (Lantmäteriet, 2024)	Safety distance of 250 m with score 0, then 1 and linear decrease to 0.01 for 3 km and beyond (P4) Safety distance of 250 m with score 0, then 1 and linear decrease to 0.01 for 30 km and beyond (P2, P3)
Electricity grid: National electricity grid lines(P3) (Svenska kraftnät, 2021)	Safety distance of 250 m with score 0, then 1 (P3)
Electricity grid: Local electricity grid lines (P3) (Lantmäteriet, 2020a)	Buffer distance of 50 m with score 0.5, outside score 1 (P3)
Stations: Stations with voltage level 130 KV (Lantmäteriet, 2020a)	Safety distance of 250 m with score 0, then 1 and linear decrease to 0.01 for 30 km and beyond (P2, P3, P4)
Stations: All other stations regardless of voltage level (Svenska kraftnät, 2021, Lantmäteriet, 2020a)	Safety distance of 250 m with score 0, then 1 (P2, P3, P4)
Roads: National and municipal roads (P2, P4) (Swedish Transport Administration, 2024)	Safety distance 250 m score 0, then score 1 and linear decrease to 0.01 for 30 km and beyond (P4) Safety distance 250 m score 0, then score 1 and linear decrease to 0.01 for 3 km and beyond (P2, P3)

Roads: Private roads (P2, P4) (Swedish Transport Administration, 2024)	Safety distance 50 m score 0, outside as 1 (P2, P3, P4)
Socio-cultural values	
Residential areas (P2, P3, P4), community functions (P4): fire station (P4), town halls (P4), schools (P3, P4), hospitals (P3, P4), prison (P4), holiday homes (P3) (Lantmäteriet, 2024)	Safety distance of 800 m score 0, outside linear increase to 1 for 1.5 km and beyond (P2, P4) Safety distance of 800 m score 0, outside linear increase to 1 from 800 to 1.2 km and stays 1 after 1.2 km (P3)
Outdoor recreation: ANI unbroken coast (P3, P4) (Länsstyrelserna, 2020)	Inside score 0, outside score 1 (P3, P4)
Outdoor recreation: ANI recreation (*P2, P3, P4), ANI outdoor recreation (P4) (SEPA, 2023a)	Inside score 0.01, outside score 1 (P3, P4) *P2: Inside score 0, outside linear increase to 1 for 1000 m and beyond
Outdoor recreation: Hiking trails (*P2, P3, P4), pilgrim trails (P4), world heritage trails (*P3) (Länsstyrelserna, 2017, Lantmäteriet, 2024)	Distance of 50 m score 0.01, outside linear increase to 1 for 300 m and beyond (P4) *P2, *P3: Distance of 50 m score 0.01, outside linear increase to 1 for 500 m and beyond
Outdoor recreation: Camping sites (P2, P3, P4), golf courses (P2, P3, P4) and allotment gardens (P2, P3, P4) (Lantmäteriet, 2024)	Inside score 0, outside linear increase to 1 for 500 m and beyond (P2, P3, P4)
Outdoor recreation: Landscape protection (P2, P3) (SEPA, 2018b)	Inside score 0, outside linear increase to 1 for 1000 m and beyond (P2) Inside score 0, outside score 1 (P3)
Outdoor recreation: Regional landscape analysis (P3-VN) (CAB Västernorrland, 2010)	Landscapes that are seemingly resilient to wind farms score 1, landscapes where the resilience is dependent on the exact position and design of the wind farm score 0.5, landscapes sensitive to wind farms score 0.1(P3-VN)
Cultural values: Culture reserves (P2, P3, P4), world heritage sites (*P2, P3, P4), landscape protection (P4) (Riksantikvarieämbetet, 2023, SEPA, 2023b, SEPA, 2022a)	Inside score 0, outside score 1 (P2, P3, P4) *P1: Inside score 0, outside linear increase to 1 for 1000 m and beyond (World heritage sites P1)
Cultural values: ANI cultural values (P1, P2, P4) (SEPA, 2024)	Inside score 0.01, outside score 1 (P2, P3, P4)
Cultural values: Ancient remnants (P2, P3-VN, P4) (Riksantikvarieämbetet, 2020)	Inside score 0.2, outside score 1 (P4) Inside score 0, outside score 1 (P2) Buffer distance of 50 m with score 0.2, outside score 1 (P3-VN)
Cultural values: Cultural environments in municipalities (P3-VG) (CAB Västra Götaland, 2019)	Inside score 0.5, outside score 1 (P3-VG)
Reindeer herding	

Reindeer herding: ANI reindeer herding (P2, P3-VN) (Sami Parliament, 2018)	Inside score 0.01, outside score 1 (P2, P3-VN)
Reindeer herding: Reindeer path (P2, P3-VN) (Sami Parliament, 2018)	Inside buffer distance of 300 m score 0.01, outside score 1 (P2) Inside score 0.01, then linear increase to 1 for 500 m and beyond (P3-VN)
Nature conservation values	
Forest: Protected forest areas (P2, P3, P4), Nature conservation agreements (P3, P4), (Biotope protection (*P2,P3,P4), (Skogsstyrelsen, 2024, SEPA, 2022c)	Inside score 0, outside score 1 (P3, P4) *P2: Inside score 0, outside linear increase to 1 for 1000 m and beyond
Forest: Key biotopes (P2, P3, P4), Nature conservation areas (P2, P3, P4) (Skogsstyrelsen, 2024)	Inside score 0.01, outside score 1(P1, P2, P4)
Forest: Forest core areas of green infrastructure (P2, P3) (CAB Västernorrland, 2020)	Inside score 0.01, outside score 1(P2, P3)
Forest: Forest connectivity zones of green infrastructure (P2,P3) (CAB Västernorrland, 2020)	Inside score 0.1, outside score 1 (P2) Inside score 0.2, outside score 1 (P3)
Wetlands: RAMSAR sites (P2,P3,P4) (SEPA, 2020)	Inside score 0, outside score 1 (P2, P3, P4)
Wetlands: Wetland inventory (*P2, P3, P4), peatland protection area (P4), Forest wetland (P4), rewetting area (P4) (SEPA, 2022b, Skogsstyrelsen, 2024)	Inside score 0.01, outside score 1 (P4) *P2: Inside score 0, outside score 1 (wetland inventory P2, P3)
Wetlands: Forest wetland (P3) (Swedish Forest Agency, 2021)	Inside score 0.01, outside score 1 (P3)
Wetlands: Wetland connectivity zones of green infrastructure (P2, P3) (CAB Västernorrland, 2020)	Inside score 0.1, outside score 1 (P2) Inside score 0.2, outside score 1 (P3)
Red throated diver: breeding sites (P2, P3-VN) (CAB, 2018)	Inside score 0.01, outside linear increase to 1 for 500 m and beyond (P2, P3-VN)
Golden eagle: suitable habitat (P2, P3-VN) (Johansson, 2020)	Habitat suitability score ranging from 0 to 1 (P2, P3-VN)
Other protected and valuable nature areas: National parks (*P2, P3, P4), nature reserves (*P2, P3, P4), nature monuments (P4), animal and plant protection areas (*P2, P3, P4), other biotope protection areas (*P2, P4), interim protection areas (P4) (SEPA, 2024)	Inside score 0, outside score 1 (P3, P4) *P2: Inside score 0, outside linear increase to 1 for 1000 m and beyond
Other protected and valuable nature areas: Natura 2000 Habitat Directive (*P2, P3, P4), Natura 2000 Birds Directive (*P2, P3, P4), ANI nature conservation (*P2, P3, P4), ANI water protection areas (P4), regulated areas with access prohibition (P4) (SEPA, 2024)	Inside score 0, outside score 1 (P3, P4) *P2: Inside score 0, outside linear increase to 1 for 1000 m and beyond
Other protected and valuable areas: Sites of high nature conservation value (P2) (CAB Västernorrland, 2018)	Inside score 0.01, outside score 1 (P2)

Other protected and valuable areas: Shore protection for water bodies (P2, P3) (CAB, 2018)	Inside buffer distance of 100 m score 0, outside score 1 (P2, P3)
Other protected and valuable nature areas: Restorable meadow and pasture land (P4) (Jordbruksverket, 2024)	Inside score 0.01, outside score 1 (P4)
Other land use and infrastructure	
Wind power exclusion areas (P3, P4): ANI for planned railways, yard, terminal or depots (P4), ANI for existing train stations, tracks, railways infrastructure, yard, terminal or depots (P2, P3, P4) (Försvarsmakten, 2024, Swedish Transport Administration, 2024)	Inside score 0, outside score 1 (P2, P3, P4) Safety distance of 250 m score 0, outside score 1(railways-P2)
Defence: Minimum safety altitude, area with special need for barrier-free access (P3), area of influence for noise (P3), weather radar and others (P3), area of importance on land (P3, P4), ANI at sea (P3, P4), ANI at land (P3, P4), exclusion zone for tall objects (P3, P4) (Försvarsmakten, 2024)	Inside score 0.5, outside score 1 (P3)
Infrastructure: Airports (P2, P3), Airport influence area (P2, P3) (Lantmäteriet, 2020)	Inside score 0, outside score 1 (P2, P3)
Infrastructure: Harbours (P2, P3) (Lantmäteriet, 2020)	Safety distance of 250 m score 0, outside score 1 (P2)
Other land-use: ANI for valuable substances (P2), Agricultural fields (P2) (CAB, 2018, SEPA, 2018a)	Inside score 0, outside score 1 (P2)

Weighting methods across different papers:

We used different weighting methods in Paper 2 and Paper 3. Paper 2 used AHP and Paper 2 used the Ranking-Rating (RR) method. The AHP method was developed by Saaty (1990) as a structured decision-making method based on pairwise comparisons. The method consists of three main steps: decomposition, comparative judgment, and synthesis of priorities (Saaty, 1990, Malczewski and Rinner, 2015).

Decomposition involves breaking down a complex decision problem into a hierarchical structure that captures its essential elements. Typically, this hierarchy includes a goal, objectives, criteria (or attributes), and alternatives. However, in this thesis, the decomposition is limited to factors and subfactors, excluding explicit representation of alternatives. This is because the analysis involves a very large number of spatial alternatives (i.e., raster pixels), making pairwise comparisons at the alternative level impractical. Instead, the hierarchy referred to here as the *planning hierarchy* structures the influencing factors and preconditions into meaningful clusters (see Figure 3).

Comparative judgment refers to the pairwise comparison of elements within each level of a hierarchy. Factors and subfactors are compared for their relative importance within each cluster. The pairwise comparison is conducted using a

"Saaty scale" ranging from 1 to 9 (Saaty, 1990). The Saaty scale is used to transform verbal judgements into numerical quantities. The Saaty scale and its interpretation are listed in Table 6.

Table 6: Saaty scale and its interpretation (Saaty, 1990)

Intensity of importance	***Definition***
1	Equal importance
3	Moderate importance
5	Strong importance
7	Very strong or demonstrated importance
9	Extreme importance
2, 4, 6, 8	Intermediate values

These comparisons are used to derive weights that reflect the relative importance of criteria. This involves constructing a pairwise comparison matrix $A = [a_{ij}]$, where each element a_{ij}represents the relative importance of criterion i over criterion j. The matrix is normalised by column sums to derive the relative priority weights. The normalisation is performed as:

$$\tilde{a}_{ij} = \frac{a_{ij}}{\sum_{i=1}^{n} a_{ij}} \qquad \text{Equation 1}$$

The priority vector (weights) w is then derived from the normalised matrix. The consistency of the judgments is evaluated using the consistency ratio (CR), which assesses the logical coherence of the pairwise comparison matrix. The consistency index (CI) is defined as:

$$CI = \frac{\lambda_{\max} - n}{n - 1} \qquad \text{Equation 2}$$

where $\lambda_{\max}$ is the maximum eigenvalue of matrix A, and n is the number of criteria. The consistency ratio is then calculated as:

$$CR = \frac{CI}{RI} \qquad \text{Equation 3}$$

where RI is the random index, which depends on matrix size. A CR value below 0.10 is generally considered acceptable, indicating satisfactory consistency in judgements. If $CR > 0.10$, the pairwise comparisons should be revised to improve consistency (Malczewski and Rinner, 2015).

The final step, synthesis of priorities, involves deriving global weights by multiplying the local weights across the hierarchy such that subfactor scores are adjusted by their respective higher-level criterion weights. These are input to the overall suitability of each spatial unit, as described in the following section.

The RR method is a somewhat simplified weighting approach based on elements from established approaches: ranking (Stillwell et al., 1981), rating (Malczewski, 1999), pairwise comparison (Saaty, 1979, Saaty, 1990), Best Worst Method (Rezaei, 2015) and weight cluster visualisation (Sjöstrand et al., 2021). It has similarities with the SMART method (Edwards, 1977), with two main differences: the use of pairwise comparisons on a 1–9 scale (Saaty, 1990), and the immediate visualisation of weights within a planning tree structure. This visual representation provides a structured overview of the resulting weights, facilitating stakeholder engagement and discussions. The compensatory nature of weights, in which decisions about the importance of factors are made only at the expense of other factors, is clearly visualised when weights are altered.

The RR method consists of three main steps (A–C):

Step A involves ranking all factors in a top-down manner, from the most to the least important.

Step B consists of rating the factors through pairwise comparisons in a bottom-up process. Each factor is compared to the least important factor, which serves as a reference. Comparisons are conducted using a scale from 1 to 9, where 1 indicates equal importance, and 9 indicates extreme importance (Saaty, 1990). The final weight w_i for each criterion is calculated as in Equation 4:

$$w_i = \frac{s_i}{\sum_{j=1}^{n} s_j} \qquad \text{Equation 4}$$

where s_i represents the value obtained from the pairwise comparison and n is the total number of criteria.

Step C involves visualising the final weights in a solar diagram structured according to the planning tree. This visualisation provides an intuitive overview of how weights are distributed across clusters, helping stakeholders understand them. It also enables stakeholders to actively engage with and adjust the weights, ensuring they reflect their preferences and encouraging iteration and discussion.

3.4.2 Design stage

General design stage

The design stage integrates spatial data, including both factors and constraints with stakeholder preferences to generate a constraint map, a suitability map, and a conflict map, which together form the basis for identifying planning alternatives. When entering this stage, each raster cell represents a potential alternative, and we need sufficient information to either eliminate or carry forward these pixels to the evaluation stage. In Paper 2, all three maps (constraint, suitability, and conflict) were used in combination to identify planning alternatives. In contrast, Papers 3 and 4 relied only on the constraint and suitability maps for this purpose. The following paragraphs describe three steps in detail, explaining how these maps were constructed and applied.

In the first step, constraint mapping was performed by aggregating all constraint maps to eliminate unsuitable areas from further analysis. This is done by multiplying all the constraint maps, that have the values 0 and 1, to create a combined constraint map. This step ensures that all infeasible or constrained areas are eliminated in subsequent steps.

In the second step, suitability mapping was performed. The factor maps were aggregated using WLC, based on stakeholder-derived weights. This produced a suitability score for each pixel on a continuous scale. A suitability cut-off threshold was then applied to identify the most suitable areas. This threshold can either be predefined or adjusted to achieve a target size, depending on the planning objective.

WLC is based on two simple assumptions. The first assumption is called linearity and concerns the weights' states: *"desirability of an additional unit of an attribute is constant for any level of that attribute"*. The second assumption of additivity means "*attributes under consideration are mutually preference independent of each other"* (Malczewski and Rinner, 2015). In this study, WLC was implemented by combining standardised factor maps through a weighted summation. Each factor map is multiplied by a weight derived from stakeholder preferences and aggregated to produce a continuous suitability score for each raster cell, see Equation 5:

$$SS(j) = \sum_{i=1}^{n} w_i \, (F_{ij}) \qquad \text{Equation 5}$$

where each weight wi is multiplied by the corresponding standardised score Fij for factor i and alternative j, and the results for all criteria are summed to obtain a suitability score SS(j) for each pixel (Malczewski and Rinner, 2015). This allows each location to be evaluated based on both its spatial characteristics and

the relative importance assigned to different factors. The suitability score is subsequently used to identify areas that meet a predefined threshold or target area requirement. Due to the compensatory nature of WLC, high suitability scores may result from trade-offs between factors. This aspect is addressed in Paper 2 by including a conflict mapping step, which is described in the following section.

In the third step of conflict mapping, disagreements between criteria within each pixel were accounted by inverse factor maps to derive a conflict score. Each factor map was inverted by subtracting its standardised value from the ideal value (1), following approaches such as Hanssen et al. (2018). The resulting maps were squared and aggregated to emphasise locations with large discrepancies between factor scores. The conflict score was calculated as in Equation 6:

$$\text{Conflict score} = \sqrt{\frac{\sum(1 - \text{standardised factor map})^2}{n}} \qquad \text{Equation 6}$$

where n is the number of factor maps. A conflict cut-off threshold was then applied to exclude areas with high internal disagreement between criteria. Similar to the suitability threshold, this cut-off can be predefined or adjusted to represent a pre-set level of conflict, or to retain a desired amount of land for further analysis.

Generation of planning alternatives

In Paper 2, planning alternatives were derived from areas identified as having high suitability and low conflict after the final filtering step in the design stage. From the resulting areas, ten planning alternatives of approximately 10 km² were identified through visual examination and digitisation, targeting spatially cohesive areas that simultaneously exhibited high suitability and low conflict.

In Paper 4, planning alternatives were derived using a more systematic spatial procedure based on the suitability map. A suitability threshold of 0.75 was applied to identify candidate areas. The resulting polygons were then processed to ensure comparable sizes:

- Small areas (< 2 km²) were excluded,
- Larger areas were subdivided using a fishnet grid (~8 km²), and
- Remaining small polygons were merged with neighbouring areas to form contiguous units.

This procedure resulted in a set of planning alternatives ranging from approximately 2 to 13 km², suitable for further evaluation.

3.4.3 Evaluation stage

Simplified evaluation in Paper 2

In the evaluation stage of Paper 2, planning alternatives of sufficient size (about 10 km²) that meet the defined thresholds for high suitability and low conflict were selected. These alternatives were then compared using selected evaluation methods, such as analysing their factor scores and mean suitability and conflict scores, to determine their final rankings.

Detailed evaluation in paper 4

Socio-ecological evaluation: The socio-ecological evaluation of planning alternatives was conducted based on six groups of factors: residential areas, cultural values, recreational values, forests, wetlands, and other protected and valuable natural areas. Planning alternatives were initially screened to identify those exhibiting ideal performance across all factors, defined as having a factor score of 1 for each criterion. These were termed ideal sites and were assigned the highest rank. The remaining alternatives were then evaluated using non-dominated sorting, a method commonly used in multi-objective optimisation to identify solutions that are not outperformed across all criteria simultaneously (Belton and Stewart, 2002b). An alternative is considered non-dominated if no other alternative performs better across all factors.

The first set of non-dominated alternatives forms the Pareto front, representing the best possible trade-offs among the socio-ecological factors (Deb et al., 2002). In this study, these alternatives were referred to as leader alternatives (or leader sites) because they represent optimal combinations of factors without requiring subjective weights or preferences. Since none of these alternatives is strictly superior to the others, selecting among them would require additional preference-based considerations. Alternatives that were dominated by these leaders were classified as followers, representing sites with similar but comparatively weaker performance. These followers were grouped based on whether they were exclusive or overlapping.

To further rank the followers, the Euclidean distance between each follower and ideal sites was calculated in a multi-dimensional factor space. This approach is consistent with the principles of the TOPSIS method, which is based on the idea that preferred alternatives should have the shortest distance to the ideal solution (Hwang and Yoon, 1981). In this representation, each alternative corresponds to a point defined by its scores across all socio-ecological factors. The Euclidean

distance thus reflects the overall difference between factor profiles, with smaller distances indicating a closer resemblance to the ideal sites.

Techno-economic evaluation: The metrics used for the techno-economic evaluation were the Levelized Cost of Energy (LCoE) and the Annual Energy Yield (AEY). LCoE represents the average net cost of producing one kilowatt-hour (kWh) of electricity over the entire lifetime of a project. It is used to compare the financial viability and competitiveness of different electricity generation projects regardless of their size, operational life, or differing capital and operational costs. It is expressed as a cost per unit of energy, typically in dollars per kWh. A lower LCoE indicates a more economically attractive project, making it a key metric for wind power developers when assessing project feasibility. LCoE depends on several factors, and it is calculated based on Equation 7:

$$LCOE = \frac{\sum_{t=1}^{n} \frac{I_t + M_t + F_t}{(1+r)^t}}{\sum_{t=1}^{n} \frac{E_t}{(1+r)^t}} \qquad \text{Equation 7}$$

where I_t is the investment expenditure (capital expenditure - CAPEX) in every year t; M_t is the operation and maintenance (O&M) expenditure every year t; F_t is the fuel expenditure (not applicable in the case of wind power) every year t; E_t is the electrical energy generated every year t, and r is the discount rate that reflects the cost of capital and risk.

The two main components of this equation are CAPEX and O&M cost. CAPEX includes the permitting process, land acquisition costs, construction of turbines and infrastructure, and other equipment and physical facilities. The assumed value is 1,070 EUR/kW based on long-term market analysis from SVK (2024). There are also additional costs, including grid connection costs (i.e., building new power lines to connect wind sites to substations) and O&M costs, which include property taxes and insurance, network costs, labour, planned and unplanned maintenance, and general upkeep of the turbines and infrastructure. The assumed value is 30 Euro per kilowatt of installed capacity (EUR/kW) based on SVK's long-term market analysis (SVK, 2024).

Grid expansion cost: In many SMCA studies on wind power, distance to the grid is typically used as the sole indicator of grid suitability. However, in this study, grid connection costs were estimated using the least-cost transmission route between potential wind farm sites and nearby substations. This was implemented using a cost-raster approach, in which each pixel represents the relative cost of constructing grid infrastructure, based on factors such as land cover, slope, and proximity to roads and existing power lines. These cost layers were normalised and combined into a single surface. For example, flat areas

close to roads were assigned lower costs, while steep or remote areas were assigned higher costs.

A many-to-many Dijkstra algorithm was then applied to identify the least-cost path between each site and the nearest substation within a defined distance. This approach identified the most cost-efficient route rather than the shortest one. The total cost of each path was calculated by summing the cost values of all cells along the route. Finally, the total distance of the required new power grid was calculated based on the least-cost path and multiplied by a unit cost of 400,000 €/km, based on the standardised asset values for 130 kV regional overhead power lines established by the Swedish Energy Markets Inspectorate (Energimarknadsinspektionen, 2023).

In addition to the power grid cost, the costs of grid connection infrastructure were considered. The required switchgear system was represented by a standard 130 kV air-insulated bay with a fixed cost of €350,000, covering circuit breakers, switches, and monitoring equipment. The cost of the main 36/130 kV transformer was estimated as a function of wind farm capacity. The installed active power was converted to apparent power using a conservative power factor of 0.8 to account for practical grid requirements. Furthermore, new on-site access roads were included using a fixed construction cost of €90,000/km, representing the heavy transport requirements associated with wind turbine installation (Trafikverket, 2010).

Annual energy yield for wind farm sites: Wind speed is the commonly used indicator for assessing wind power potential (Rediske et al., 2021, Spyridonidou and Vagiona, 2020). However, the actual electricity generated by a turbine depends on more than just wind speed. Key factors include air density, the size of the turbine rotor, the turbine's power curve, and the frequency of different wind speeds throughout the year. In other words, turbine output is determined by both local physical conditions and the frequency with which those conditions occur. The annual energy yield (AEY) of a turbine is calculated as:

$$\text{AEY} = 8{,}760\ \text{hrs} \int_0^\infty P_w\,(U, \rho)\, f(U)\, dU \qquad \text{Equation 8}$$

where 8,760 is the number of hours in a year, $P_w(U, \rho)$ is the turbine power output at wind speed U adjusted for local air density ρ, and $f(U)$ is the frequency (probability) of wind speeds U occurring during the year. The turbine power output at each wind speed is obtained from its **power curve**, which shows how electricity production increases with wind speed up to a rated maximum and decreases when wind speeds are too low or too high for safe operation. To account for differences in air density, wind speeds are adjusted using:

$$U = U_{st}\left(\frac{\rho_{st}}{\rho}\right)^{z(U)} \qquad \text{Equation 9}$$

where U_{st} is the standard wind speed from the power curve, ρ_{st} is the standard air density for that curve, ρ is the actual air density, and $z(U)$ is a correction factor that varies with wind speed, see Equations 10 a)-c):

a) $z(U) = 1/3$ for $U \leq 8\text{m/s}$

b) $z(U) = 1/3 + \frac{1}{3}\frac{U-8}{5}$ for $8 < U < 13\text{m/s}$ Equations 10 a)-c)

c) $z(U) = 2/3$ for $U \geq 13\text{m/s}$

Because wind speed varies over time, they are modelled using a Weibull distribution, which describes the likelihood of different wind speeds occurring throughout the year, see Equation 11:

$$f(U) = \frac{k}{\alpha}\left(\frac{U}{\alpha}\right)^{k-1} e^{-(U/\alpha)^k} \qquad \text{Equation 11}$$

Here, k is the shape parameter (describing how spread-out wind speeds are) and α is the scale parameter (related to the average wind speed). These parameters can be obtained from wind atlases, such as the European Wind Atlas (as in this study (NEWA, 2025), or from local meteorological data. Once known, $f(U)$ can be calculated for all wind speeds at a site. By combining the turbine power curve with the wind speed frequency, it is possible to estimate the annual energy yield at a given location. This provides a robust measure of a turbine's potential electricity production, allowing planners and researchers to compare sites and integrate wind power potential into broader assessments of energy infrastructure, policy, and sustainability.

4 Results

4.1 Results and discussion – Paper 1

4.1.1 Theme 1: Year and country of study

A total of 68 papers were identified through the systematic literature review. The publication trend shows a strong increase between 2014 and 2019, with 2019 having the maximum number of publications (11). This is followed by a decline in 2020 and another peak of publications in 2024 (11). Overall, the trend is not steadily increasing, despite the continuous growth of wind energy development during the same period.

Geographically, only 11 European countries published studies on the application of SMCA for wind farm planning. Turkey ranked highest in terms of the number of publications, followed by Greece. In contrast, countries such as Germany, which have the highest levels of wind energy generation, contributed relatively few studies. Conversely, Turkey, despite having the highest number of publications, generates less wind energy than several other European countries.

This mismatch between research output and wind energy development suggests that SMCA is either not used as the methodological approach for the wind farm siting or is less reported in the scientific literature

4.1.2 Theme 2: Criteria

Criteria identified in the reviewed papers were classified into four groups: techno-economic, socio-cultural, ecological, and others. Table 7 summarises the frequency with which individual criteria were used across the reviewed studies. Among these, techno-economic criteria such as wind speed (57), power grids (55), roads (55), and slope (46) appear in many studies, indicating that SMCA publications on wind power siting are still primarily driven by technical feasibility and infrastructure considerations rather than socio-cultural or ecological perspectives.

Socio-cultural criteria, particularly distance to settlements (38) and urban areas (34), are also widely used. However, social impacts are largely represented by simple distance-based proxies, while more complex dimensions such as social acceptance, justice, and participation, although occasionally discussed, remain underrepresented in practice. Other criteria, including electricity demand, land ownership, and compatibility with existing planning, are rarely considered, despite their importance in real-world decision-making.

Table 7. Summary of criteria applied in reviewed papers.

Techno-economic	Socio-cultural	Ecological	Other
Wind speed/potential (57) Power grids (55) Roads (55) Slope (46) Rails (24) Elevation (23) Transformer stations (8) Aspect (8) Terrain Roughness (7) Wind density (7) Grid capacity/voltages (6) Wind direction (2) Capacity factor (2)	Residential/Settlements (38) Urban life (34) Incompatible land uses (24) Culture/Heritage sites (20) Irrigated/Agricultural land (19) Scenic landscapes (19) Archaeological/Historical sites (15) Outdoor recreation/Recreation centres (12) Tourism areas (12) Social acceptance (12) Existing turbines/Other RES (10) Noise (10) Visibility (9) Churches, monasteries (6) Labour share/Local employment (5) Livestock (3) Population (3) Community interest sites (3) Age, qualification (2) Electricity demand (2) Reindeer herding (2) Land ownership (2) Land cost (2) Land availability (2) Compatibility with existing planning (1)	Protected areas (42) Water bodies (34) Forest (34) Wetlands (21) Birds (19) Bats (5) Ecological corridors (2) Vegetation index (2) Mountain (1) Ravines (1) Other endangered species (1)	Airports (27) Radar/Antenna/Radio mast/ Telecommunications (15) Landslide/Flood risk/Seismic prone (13) Mines (11) Coastline (11) Military (10) Fault lines (10) Soil type/Rocky land/Geology (7) Ports/Harbours (4) Sand dunes (4) Construction sites (2) Dumps (1)

In a few studies, more advanced approaches such as visibility analysis or noise modelling are included. Existing wind turbines are also used in multiple ways: as constraints, as indicators of social acceptance, and as a validation tool for suitability models. Additionally, land uses related to recreation and tourism are sometimes included to capture competing land-use interests.

Within the ecological category, protected areas (42) are the most frequently applied criterion, whereas more detailed biodiversity-related aspects, such as birds, bats, and species-specific considerations, are included in relatively few studies.

Overall, the review highlights a clear tendency to rely on readily available and easily quantifiable spatial data. In contrast, more context-specific, complex socio-cultural and ecological factors remain underrepresented.

4.1.3 Theme 3: Weighting method and decision hierarchy

Weighting exercise is influenced by stakeholder perspectives, which do not aim for consensus but rather for transparent inclusion of different viewpoints (Hobbs and Meier, 2000).

Decision hierarchy

Decision hierarchy provides an overview of the decision problem by clustering the main goal into criteria and sub-criteria, thereby promoting shared understanding and discussion (Belton and Stewart, 2002a). Only 19 of 68 papers used an explicit decision hierarchy, with the majority of studies having 3 to 5 clusters. The common clusters are environmental/ecological, technical, social, and economic. The same criterion can be placed in different clusters in different studies, meaning there is no strict basis for creating these clusters (e.g. wind speed under technical versus environmental) (Bilgili et al., 2024, Yaman, 2024).

Weighting methods

Weighting was applied in 53 out of 68 papers, with AHP (32) being the most frequently used method, followed by fuzzy AHP (8). Several advantages of AHP reported in the reviewed papers include its ability to structure complex decision problems into a hierarchy, its use of intuitive pairwise comparisons, and the inclusion of consistency checks to evaluate the reliability of judgments (Latinopoulos and Kechagia, 2015, Sotiropoulou et al., 2024, Szurek et al., 2014, Watson and Hudson, 2015). ANP, DEMTAEL and Best-Worst Method (BWM) were also observed to be applied in the papers.

Alternative approaches, such as exclusion-based GIS screening, SEA-based evaluation, optimisation models (e.g. NSGA), and regression-based suitability

modelling, were also applied instead of explicit weighting. While these methods can reduce reliance on subjective weights, they may also obscure trade-offs and reduce transparency in the decision-making process. To address this limitation, some studies incorporated sensitivity analysis (Ioannidis and Vagiona, 2024), conflict scoring (Manolan Kandy et al., 2024), or alternative aggregation methods to better reveal trade-offs and improve transparency in the results.

4.1.4 Theme 4: Sensitivity analysis

Sensitivity analysis was conducted in 31 out of 68 papers, primarily to identify robust solutions and increase confidence in the results for planners and stakeholders. This is typically done by testing the effects of changes in criteria, weights, and methodological choices.

Altering the criteria by including or excluding factors, or by modifying the standardisation, can have a stronger influence on results than changes in weights. In some studies, different threshold settings or policy scenarios were also tested, showing how regulatory assumptions can significantly affect spatial outcomes.

The most common form of sensitivity analysis is weight variation. This includes equal-weight scenarios, stakeholder-based weighting schemes, and alternative weighting scenarios such as environmental versus socio-economic prioritisation. The results show mixed stability: in some cases, rankings remain robust, while in others they are highly sensitive to changes in weights.

Finally, several studies compared different MCDA methods (e.g. TOPSIS, VIKOR, COPRAS, fuzzy AHP). While results are often broadly similar, these comparisons can still lead to changes in rankings or differences in the areas identified as suitable, highlighting the influence of methodological choices.

4.1.5 Theme 5: Scale of the study area

Of the 68 reviewed papers, 59% focused exclusively on design problems (site identification), 16% on evaluation problems (ranking predefined alternatives), and 25% combined both by first identifying potential sites and then evaluating them. Study areas vary widely (from very local, from 18 km2 for a district to 78,000 km2 to national scale) in the reviewed papers. These reflect the different administrative hierarchies observed in various papers. Thirteen papers from countries such as the Netherlands, Greece, Poland, Germany, and Sweden reported national-scale studies. There were 22 papers from the second level of the administrative hierarchy, where organisational units differ between countries. Fewer papers had study areas at the third administrative level (9 papers). The choice of scale impacts which criteria are relevant, how criteria are standardised and how stakeholders' preferences are interpreted. During the

standardisation step, all criteria must be transformed into a common suitability scale, referred to as a factor map.

Standardisation of criteria can follow two main approaches: global and local scaling. Global scaling applies a single value function or treatment uniformly across the entire study area, resulting in homogeneous standardisation. In contrast, local scaling allows the standardisation of a criterion to vary within the study area, reflecting spatial differences in different conditions. For example, wind speed could be standardised using a linear function in flat inland areas, while a non-linear (e.g., fuzzy or convex) function could be applied in coastal areas to reflect differing perceptions of suitability.

Global scaling is most observed in the reviewed studies. In these cases, thresholds, suitability functions, or classes are often borrowed from existing literature, leading to the use of generic value scales. For instance, wind speed thresholds may be derived from turbine specifications or financial feasibility, and minimum distances to residential areas are typically based on safety standards or regulatory guidelines.

In other cases, standardisation is informed by stakeholder input or specific characteristics of the study area, resulting in context-specific thresholds and value functions. For example, wind speed may be scaled relative to the range observed within the study area rather than using externally defined thresholds. While such approaches better reflect local conditions, they may also reduce the transferability of results, as what is considered optimal locally may not be optimal in a broader regional or national context. This could also be the case when standardisation is used while evaluating the predefined alternatives.

It is therefore important to ensure consistency in the standardisation approach. Mixing generic (externally derived) and context-specific (internally derived) value scales can complicate weighting and aggregation, potentially influencing the resulting suitability maps.

In other words, standardisation varies not only in how it is applied spatially (global versus local scaling), but also in the origin of thresholds (generic versus context-specific). These distinctions have important implications for comparability, transferability, and the overall robustness.

Most reviewed studies are conducted at regional or national scales, reflecting growing interest in strategic planning approaches to support a more balanced distribution of wind farms. Through this, a more balanced sharing of burdens and benefits, based on energy supply and demand, can also be achieved.

However, large-scale planning introduces several challenges. The selection of criteria becomes more complex and broader, as factors relevant in one area may not be applicable in another, and stakeholder representation becomes more difficult to define. In contrast, small-scale planning allows for more detailed

analysis and facilitates the inclusion of local knowledge, stakeholder input, and context-specific conditions.

These scale-related dynamics are not only analytical and technical but also relate to how spatial decision-making in wind power planning is organised and reflected in governance structures. As shown by González et al. (2016), decentralised decision-making can reduce opposition to wind power development but may also lead to fragmented implementation, with developments unevenly distributed across regions.

A similar situation with wind farm distributions can be observed in Sweden, where strong municipal autonomy exists, including the municipal veto. While national strategies define indicative wind energy targets at the county level, these are not legally binding, and local authorities retain control over siting decisions. This multi-level governance structure enhances local legitimacy but also creates coordination challenges between national ambitions and local priorities (Wretling et al., 2022).

These findings underscore the importance of integrating top-down and bottom-up perspectives in wind power planning. Large-scale, top-down approaches can provide strategic direction and support long-term system optimisation (Sliz-Szkliniarz, 2019, Santana-Sarmiento et al., 2019), while small-scale, bottom-up analyses offer detailed insights, improve transparency, and strengthen stakeholder engagement.

4.1.6 Theme 6: Stakeholder involvement

Stakeholder participation in the reviewed studies was through questionnaires, interviews, focus group discussions, and workshops. In 59% of the papers, stakeholders or external experts beyond the author group were involved. These participants typically included academics, energy experts, government representatives, planners, and local actors with experience in wind energy. The number of stakeholders varied considerably, ranging from 1 to 22 individuals per study.

Stakeholders were engaged at various stages of the SMCA process, though involvement was most frequent during the weighting exercise. Participation in earlier stages, such as criteria identification and the development of standardisation procedures, was less common. In several cases, however, the reporting of stakeholder involvement lacked sufficient detail, making it difficult to clearly assess their exact roles and level of influence.

Involvement in problem structuring was particularly rare. Overall, the definition and structuring of criteria were predominantly expert-driven rather than stakeholder-driven across most studies. This aligns with the distinction proposed by Munda (2008) who argues that stakeholders should primarily provide values, preferences, and contextual input, whereas technical tasks such

as formal problem structuring, consistency checking (e.g., non-redundancy and clarity), and methodological formulation should be handled by experts.

Some studies, however, highlight a stronger role for stakeholder input in shaping criteria and problem framing. For example, (Manolan Kandy et al., 2024, Stoltmann, 2020) demonstrated stakeholder involvement across multiple stages of the SMCA process, including earlier phases such as criteria development and structuring.

Finally, several papers emphasised the importance of public acceptance in SMCA applications. In these cases, social acceptance was sometimes explicitly included as a criterion. Stakeholder input was typically collected through surveys and questionnaires, with the interpretation of public views often left to experts rather than being directly integrated into the decision-making process.

4.1.7 Theme 7: Real-world planning uptake

The reviewed papers commonly begin by emphasising the importance of expanding renewable energy, with a particular focus on wind energy. Wind energy development is strongly driven by ambitious European and national climate and energy targets, which are frequently used to justify the application of SMCA in wind farm siting (Siyal et al., 2016, Latinopoulos and Kechagia, 2015, Ifkirne et al., 2022).

However, despite this policy-driven motivation, the reviewed literature shows variation in the extent to which SMCA studies engage with real-world planning. Only two papers (3%) were reported to be part of an actual planning project. In total, 54 papers addressed planning: 5 only briefly mentioned it, while 49 discussed its applicability. In contrast, 14 papers focused on methodological aspects or other topics not directly linked to planning.

Three key challenges discussed in the reviewed papers regarding real-world wind power planning are social acceptance, distributive justice, and power monopolies. Local social acceptance is a major reason for the rejection of wind power projects (Harper et al., 2019, Latinopoulos and Kechagia, 2015), despite techno-economic feasibility often being well understood. In contrast to technical factors, social dimensions are more complex and typically represented in SMCA studies through indirect socio-economic proxies such as distance to dwellings, recreational areas, noise and visual impact models, or demographic variables (e.g. age, education, political orientation). However, these indicators only provide a fragmented and indirect representation of actual local concerns.

Only a limited number of studies integrate participatory approaches or directly include public opinion in wind farm siting (Ioannidis and Vagiona, 2024). Examples include survey-based approaches (Josimović et al., 2023) and qualitative assessments of local sentiment (e.g. through expert judgement or media analysis) (Bili and Vagiona, 2018). This lack of integration is largely

explained by methodological challenges, including high data collection effort, uncertainty, and limited transferability between contexts (Tsani et al., 2024).

Social acceptance can be understood through three dimensions: socio-political acceptance, market acceptance, and community acceptance (Wüstenhagen et al., 2007). Among these, community acceptance—referring to local perceptions of specific projects—is the least represented in SMCA applications. Evidence from Harper et al. (2019) also suggests that broader social and demographic factors (e.g. education levels, age structure, political composition) can be more influential in project approval than commonly used spatial distance-based indicators, highlighting limitations in current proxy-based modelling approaches.

Recent research, therefore, calls for a more explicit and nuanced treatment of social acceptance in SMCA, including better representation of intangible factors such as place attachment and ecological values, as well as greater transparency regarding missing or uncertain social data (Klok et al., 2023).

Within the broader energy justice framework by Jenkins et al. (2016), wind energy planning is shaped by both distributive and procedural justice. Distributive justice concerns how costs and benefits are spatially and socially distributed, while procedural justice relates to fairness in decision-making and stakeholder participation. These issues are particularly relevant for wind power, as local communities bear long-term landscape impacts while benefits are more widely distributed across society and time, creating a clear spatial and temporal mismatch.

4.2 Results and discussion – Paper 2

Paper 2 developed the REWIND framework, a GIS-based SMCA approach structured into the scoping, design, and evaluation stages. A novel conflict mapping component was introduced to complement conventional suitability analysis. This framework was applied to Västernorrland County.

4.2.1 Key results from Paper 2

The results from Paper 2 show how restrictive the wind farm siting can be, once constraints are applied. Already in the first step, 70% of the total land area was excluded, highlighting that wind power siting is heavily constrained by legal requirements and current thresholds. Among the municipalities, Kramfors experienced the largest reduction, with 84% of its land restricted, while Örnsköldsvik retained the largest absolute area (2138.6 km^2) after consideration of the constraints.

The main reason for land constraints was socio-cultural criteria (45.8%), particularly residential areas and the 800 m buffer distance applied. This is

notable given that the region has one of the lowest population densities in Sweden, yet settlement structures play a decisive role in limiting available land. The second ranking factor for constraints concerned Ecological values, while the third and fourth were the factors Wind power development and Other infrastructure.

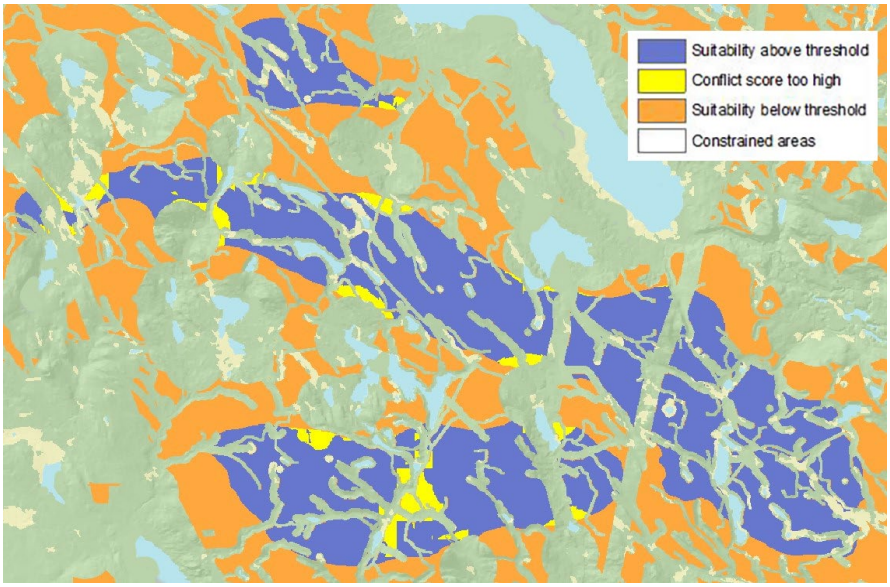


Figure 3. Illustration of the three different filtering steps: constraints removed, high-potential areas identified (blue), and conflict areas removed from the latter (yellow). The remaining blue areas have high potential for wind power development. The map is fictive.

The weighting of factors using AHP was carried out, further illustrating the importance of stakeholder influence in the analysis. The highest weights were assigned to wind resource and reindeer husbandry, whereas wetlands and red-throated diver received the lowest weights. This reflects both the composition of the stakeholders involved and their perceptions, as well as how the planning tree was structured, ultimately shaping the outcome of the suitability assessment.

Suitability scores were generated using WLC and ranged from 0 to 0.99 across the study area. Applying a suitability threshold of 0.85 reduced the available area to 2855 km². In parallel, conflict scores were calculated without weights to capture internal inconsistencies between factors. Applying a conflict threshold of 0.15 further reduced the available land to 963 km², representing areas of high suitability and low conflict (HS–LC). The largest HS-LC area was in Örnsköldsvik municipality (270 km²). An illustration of high suitability, high conflict, low suitability and constraints is shown in Figure 4.

Only a small fraction of Västernorrland retained potential at this stage, and its distribution varied significantly across municipalities. Örnsköldsvik (270 km²), Sundsvall (189 km²), Sollefteå (186 km²), and Ånge (153 km²) retained relatively large areas, while Timrå (49 km²), Härnösand (53 km²), and Kramfors (63 km²) had much less land available. This illustrates clear spatial heterogeneity, in which the same methodology produced very different outcomes depending on local conditions. More densely populated areas with high energy demands usually have less land with potential for building wind farms.

The relationships between suitability and conflict scores provide important insights into the role of trade-offs. While most high-conflict areas corresponded to low suitability (LS–HC), a substantial area (1890 km²) was classified as high suitability–high conflict (HS–HC). These areas illustrate the compensatory nature of the WLC approach, in which high scores in heavily weighted factors can offset very low scores in others. In this case, factors contributing 24% of the total weight had moderate-to-high scores (0.30 and 0.67), while several other factors had scores below 0.1.

For example, in parts of Örnsköldsvik municipality, HS–HC areas showed low scores for wetlands (0.10), grid proximity (0.20), and roads (0.30), yet still achieved high overall suitability due to strong performance in dominant factors. This demonstrates how weighting can drive strong compensation effects, allowing a few influential criteria to outweigh multiple poorly performing ones.

The conflict score addresses this limitation by identifying and excluding such areas in the final filtering step. As a result, the remaining HS–LC areas show a more balanced performance across criteria, with minimum factor scores of around 0.3 for roads and ≥0.45 for other factors. This indicates a reduction in extreme trade-offs and a more internally consistent set of planning alternatives.

4.2.2 Key data and methodological insights

Data insights

A total of 14 site-specific criteria were considered in the study, constructed from 40 spatial datasets (See Table 5). The selection of criteria was based on scientific literature, legal requirements, and expert judgement. However, this stage was associated with uncertainties and data limitations. One example is the oversimplification of the power grid factor, which is based solely on distance. This overlooks critical factors such as thermal limits, voltage conditions, and system stability, all of which significantly influence transmission capacity (Yusoff et al., 2017). In practise, new connections to wind farms cannot be established without improving the current capacity in many cases. This means that power grid factors should be treated more carefully in real-world planning.

However, the difficulty in obtaining such detailed data on power grids made it necessary to consider only distances as an initial proxy. In this study, the regional grid was considered, typically operating at 40–130 kV, to meet the needs of wind parks with around 25 MW of installed capacity.

Other datasets, such as those representing wind speed and reindeer herding, may also be subject to uncertainty regarding how well they reflect actual conditions in the study area. In the case of wind speed, MIUU data (Bergström and Söderberg, 2012) were used, which are modelled rather than measured. As such, their accuracy can vary depending on terrain complexity, model assumptions, and spatial resolution. Similarly, reindeer husbandry data from the Sámi Parliament (Sami Parliament, 2018) provide an important planning basis but represent generalised land use under a "normal year" and may not fully capture temporal variability or local conditions.

Methodological insights

According to the assumptions of WLC, it is important that factors are independent to avoid double-counting (Malczewski and Rinner, 2015). Correlations between factors can be evaluated using several methods, such as the Pearson correlation coefficient or similar approaches (Lindén et al., 2021). Where strong correlations are identified, factors can either be excluded or combined into a composite index. In this study, several overlapping factors were merged into single indicators through mosaicking. However, some degree of interdependency may remain and should be further examined.

Weights can vary significantly depending on the methodological approach, including stakeholder selection, weighting method, weight aggregation, and the structure of the planning tree. The use of an arithmetic mean to aggregate weights does not represent a negotiated consensus, which would be necessary in real-world multi-stakeholder planning. This limitation is present in the current study.

The structure of the planning tree can also influence results. For example, the category "reindeer husbandry", represented by a single factor, received the highest individual weight across all weighting profiles. This suggests that structurally simple branches may be amplified in the weighting process.

A sensitivity analysis further demonstrated the influence of weights. When the wind resource factor was assigned a weight of 0.49, the suitable area decreased to 2282.8 km² after the suitability step and to 760 km² after applying the conflict threshold. In this case, areas with very high wind scores dominated the suitability outcome, reducing the influence of other factors. This highlights how weighting can introduce "invisible trade-offs" and reinforces the need for careful interpretation and transparency in the analysis.

4.3 Results and discussion – Paper 3

Paper 3 focused on the development and application of the RR method for factor weighting within the REWIND framework.

4.3.1 Weighting using the RR method

In the Västernorrland case study, six weighting profiles were obtained as shown in Figure 5. Factor weights ranged from 0.01 to 0.25, indicating no extreme prioritizations across profiles. The wind power development cluster generally received lower weights, except for the energy and climate strategy profile, where it was more prominent. Residential areas were assigned the highest weight in three profiles and medium-to-high weights in the remaining profiles, highlighting their consistent importance. Reindeer husbandry received the highest weight in two profiles, while in the others, its importance varied considerably, ranging from low to high. Several factors reached at least a high weight in one or more profiles: Wind resource (1), Residential areas (3), Outdoor recreation (1), Cultural values (1), Reindeer husbandry (2), and Defence (1). Factors such as Power grids, Roads, Sensitive birds, and Wetlands were never assigned the highest weight in any profile.

In the Västra Götaland case study, 11 weighting profiles were obtained, with weights ranging from 0.01 to 0.2 (see Figure 5). In contrast to Västernorrland, Wind resource was the most frequently prioritised factor, receiving the highest weight in four profiles and very low weights in others, reflecting greater divergence among the stakeholders. Residential areas were the second most frequently prioritised factor, receiving the highest weight in three profiles. This factor also showed relatively consistent importance across stakeholders and study areas, making it a robust factor in the weighting process. Cultural values and Outdoor recreation were not ranked highest in any profile. Weight distributions were visualised as solar diagrams during the weighting exercise as in Figure 6. These diagrams reflect how weights are distributed across clusters in the planning tree.

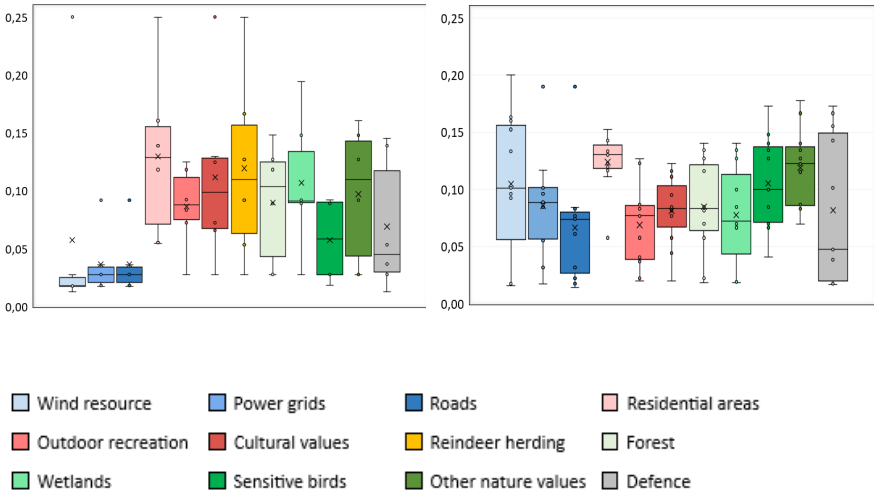


Figure 4. Distribution of weights from Västernorrland (left, N = 6) and Västra Götaland (right, N = 11) counties obtained from stakeholders through the RR method. The diagramme shows mean (x), median, and quartiles of the respective weights.

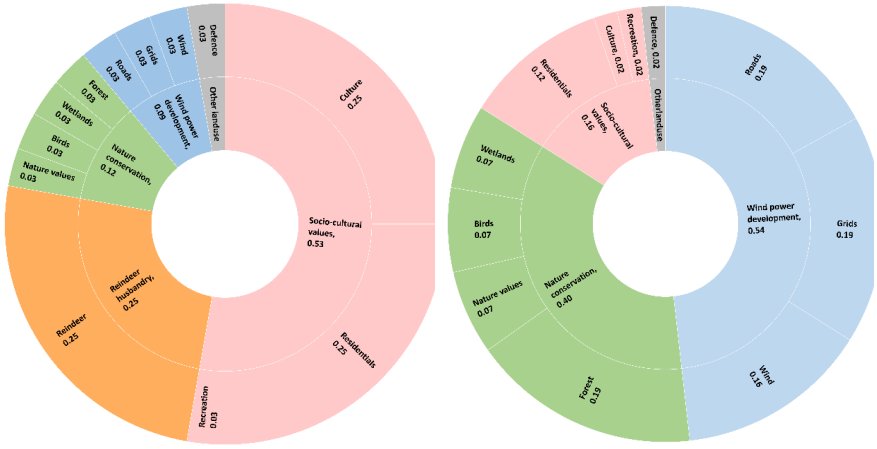


Figure 5. Examples of weight visualisations from Västernorrland (left, weight profile 1.1) and Västra Götaland (right, weight profile 2.2) counties, structured according to the planning tree.

4.3.2 Focus group interviews

Key themes emerging from the focus group interviews included: (a) selection of criteria (what to include or exclude in the analysis); (b) scaling and standardisation of spatial indicators; (c) control and interpretation of weights; (d) the role of the planning tree in structuring decisions; (e) the influence of weights on planning outcomes, particularly in the design of alternatives; and (f) stakeholder representation in the process.

Selection of criteria

This is a complex, non-trivial process that involves a wide range of criteria and spatial datasets. It is also highly dependent on the scale and location of the study area, where a criterion may be relevant in one context but not in another. Common challenges include outdated datasets, low spatial resolution, and various sources of uncertainty, for example, in wind-speed and reindeer-husbandry datasets. Another challenge lies in the major knowledge gaps regarding the detailed impacts of wind power installations on the different factors.

The need for expert involvement from relevant domains was emphasised, particularly for ensuring correct interpretation and understanding of data limitations. In some cases, data was classified and not available for analysis, such as grid capacity information. Wind power planning and grid development are strongly interdependent, highlighting the need for cross-sectoral collaboration. Participants also discussed the difficulty of translating certain factors into spatial indicators concerning socio-economic issues, particularly qualitative aspects such as social acceptance.

Scaling of spatial indicators

The scaling and standardisation of spatial indicators emerged as another complex issue in the discussions. Through standardisation, diverse spatial datasets are transformed into a single suitability scale, making the choice of scale endpoints both critical and non-trivial. According to Munda (2008), standardisation of spatial indicators is a task for the researchers, though with input from stakeholders, to minimise bias. These thresholds often depend on the level of analysis—national, regional, or local, which can significantly influence the resulting factor maps. For example, the same wind speed dataset may produce different suitability outcomes depending on whether it is standardised at a national or regional scale. This highlights the importance of clearly communicating standardisation rules to participants to ensure that factor maps are properly understood. Providing factor maps, along with explanations, prior to the weighting exercise can improve transparency and trust in the SMCA process.

The issue becomes even more complex in the case of reindeer husbandry, where suitability varied across municipalities in Västernorrland. Additionally, the spatial extent of Sami villages, which often spans multiple municipalities and counties, complicates the definition of the study's boundaries and the scaling procedures. Given these challenges, the involvement of domain experts is essential to ensure meaningful representation of factors. Furthermore, improved collaboration between institutions such as CAB, regional authorities, SVK, grid operators, and the STA is necessary to enhance data sharing and integrate expert knowledge into the planning processes.

Even when conducted carefully, standardisation inevitably introduces biases when translating technical, economic, social, and environmental criteria into a unified suitability scale. Current practices often rely on standardisation approaches borrowed from previous studies, which could be grounded in expert judgement, while some incorporate new expert input. Due to uncertainties in decision rules, the scaling process requires clear documentation, expert involvement, transparency in assumptions, and, ideally, sensitivity analyses to ensure robustness.

Control of weights

In this paper, the RR method was applied to obtain weights. The method was perceived as easy and straightforward to use, and no technical or pedagogical issues occurred during the Excel-based weighting exercise. A key question was whether the resulting weights reflected stakeholders' preferences regarding individual factors. The two-step approach—ranking from the top down and conducting pairwise comparisons from the bottom up—was perceived as reasonable and logical. The method also offers lower complexity than AHP and provides a clear overview of clusters through the planning tree visualisation.

One identified limitation was the restricted rating scale (1–9). Some participants suggested improving scale resolution by allowing more than nine steps. Another proposed improvement was to introduce factor grouping into high-, medium-, and low-importance levels, followed by comparisons within each group.

The AHP method was also discussed, as it had previously been applied with participants in Västernorrland County and is commonly used in wind power planning. In AHP, all factors are compared pairwise. It was noted that the large number of pairwise comparisons can lead to inconsistencies. Furthermore, the consistency ratio requirement (≤ 0.1) must be met by the participants. This may shift attention away from the actual preferences, and iterative adjustments to meet this threshold can make the results feel less meaningful (Asadabadi et al., 2019). The use of AHP priority tools may also influence or override participants' intended judgments. In other words, complex procedures may produce final

weights that do not fully reflect the participant's true perceptions, thereby reducing trust and the perceived meaningfulness of the process.

The planning tree

The planning tree was primarily used as a structuring device to organise factors into clusters and to provide a clearer overview of the impacts and conditions influencing wind power planning. This hierarchical structure aligns with (Saaty, 1990), who, building on (Miller, 1956), noted that cognitive capacity is limited when dealing with more than 7–9 elements at a time.We applied a stakeholder-oriented structure for the planning tree. The basis was legislation, the SDGs, and the organisation of governmental bodies and municipalities, including the Sami villages that govern reindeer husbandry. This structure was considered logical and helped identify and connect with relevant experts, thereby improving practical usability.

However, it resulted in an asymmetric planning tree, with some clusters containing many factors while one contained only a single factor. This raised a risk of overemphasising individual factors during the weighting process. The solar diagrams used during the weighting exercise helped participants visualise the weight distribution and adjust their preferences immediately. Since the weighting was compensatory and constrained to a restricted scale, increasing the weight of one factor necessarily reduced the weights of others. This applied both at the factor and cluster levels. The visualisation of the overall weight distribution within the planning tree was therefore considered useful and pedagogical.

Effect of weights on the design of planning alternatives

Planning alternatives derived from different grid scenarios influenced both the wind farm's potential and its possible size. Defence-related factors strongly affected the results, excluding several otherwise suitable areas, and were unevenly distributed across municipalities. The coexistence between wind farms and defence interests was considered difficult to plan and requires further investigation, inter-sectoral collaboration and negotiation.

Further discussions focused on suitability maps based on the average weight scenario. Some sites showed robust high suitability across weight variations, while others were highly sensitive to changes in weights. Weighting can influence outcomes through trade-offs between factors, creating a risk of hidden biases, as the role of weights may not be clearly perceived by the participants. This highlighted the need for improved visualisation of trade-offs and concerns about information loss when multiple factors are aggregated into a single suitability score.

Stakeholders' representation

Stakeholder weighting is inherently subjective, even among professionals, as individuals often tend to prioritise their own knowledge domain. This raises questions about who should participate in the weighting process and how expertise should be defined. We discussed whether participation should be limited to domain experts or extended to a broader group of stakeholders, including government representatives and local communities. A key concern highlighted in the discussions was the need for clearer guidance on handling difficult trade-offs to support robust planning decisions.

The literature reflects this variation in practice. In some studies, stakeholders include EIA project managers, planners, and technical wind farm experts, while in others, government representatives and local communities were also involved (Harper et al., 2019, Watson and Hudson, 2015). The stage at which stakeholders are engaged also varies across studies (Baban and Parry, 2001, Bili and Vagiona, 2018). Overall, real-world SMCA applications increasingly emphasise continuous stakeholder engagement throughout the entire process—including criteria selection, standardisation, and weighting—to ensure relevance, transparency, and legitimacy of the results.

4.4 Result and discussion - Paper 4

A total of 896 potential wind farm sites (see Figure 7) were identified across seven counties within high-suitability areas. The sites range in size from 2 to 13 km², with a combined area of 4,457 km². First, a comparison was made with the NWS estimates using the sites identified during the design stage. Among the seven regions, Gävleborg stands out with substantially higher electricity production potential than all other counties. Uppsala and Västmanland also show relatively high wind energy potential within SE3, although Uppsala's overall electricity production is largely driven by nuclear power rather than wind energy. In contrast, Stockholm and Södermanland exhibit low wind energy potential and currently generate limited wind power, reflecting both spatial constraints and high land-use competition, particularly in Stockholm. Örebro and Östergötland show intermediate levels of wind energy potential compared to the other regions.

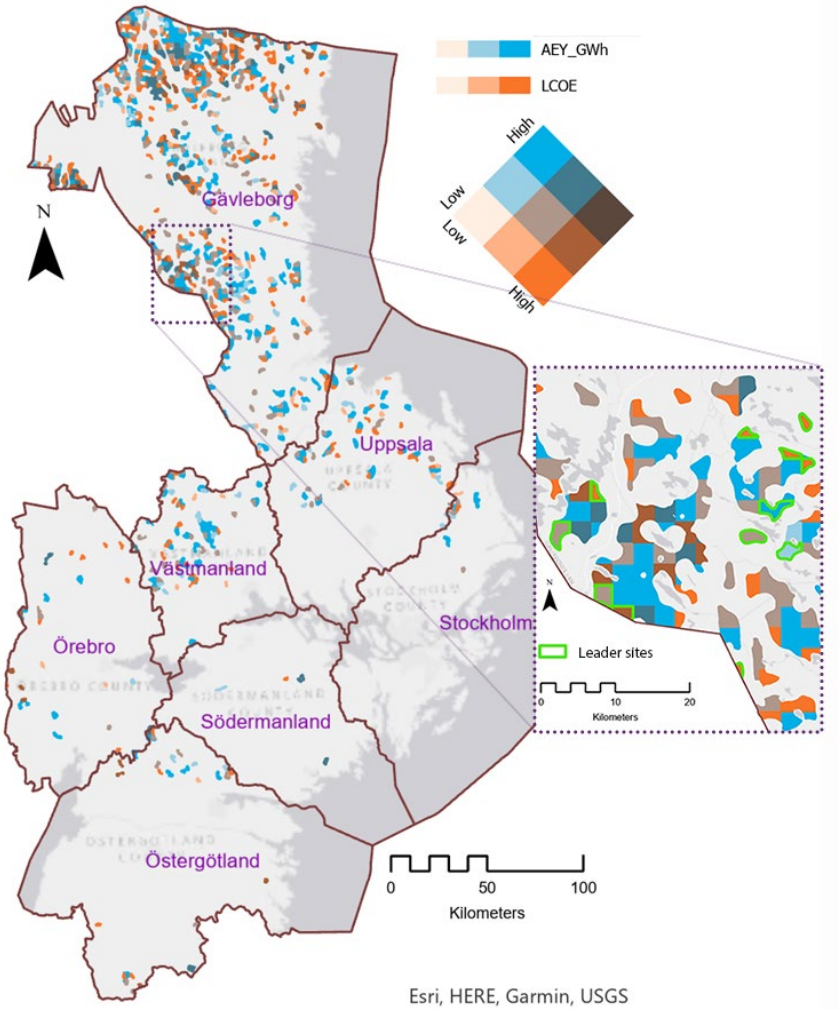


Figure 6. Annual Energy Yield (AEY, kWh), and Levelized Cost of Energy (LCoE, EUR/kWh) in all seven counties. Leader sites are highlighted in the zoomed-in window.

More sites were found in Gävleborg and the northern parts of Uppsala and Västmanland, while fewer were found further south, with the fewest in Södermanland. This pattern is likely explained by lower wind speeds and higher residential density in Södermanland. Blue areas in Figure 5 indicate high AEY and low LCoE, making them the most attractive for developers. Larger sites seem to have higher AEY and lower LCoE. AEY values range from 30.5–250.6 GWh/year, and LCoE ranges from 0.04–0.14 EUR/kWh.

No relationship was found between site size and socio-ecological ranking. Similarly, there is no correlation between socio-ecological scores and techno-economic performance scores. A subset of top-performing sites was identified based on combined criteria, including high AEY, low socio-ecological distance, and low LCoE (< 0.05 EUR/kWh). These top-performing sites were distributed as follows: 18 in Gävleborg, 2 in Uppsala, 2 in Västmanland, 1 in Stockholm, and 1 in Östergötland.

4.4.1 Gävleborg

There were 658 sites identified in Gävleborg County, covering 3346 km². Gävleborg currently produces more energy than it consumes, which is typical for the northern Swedish price zones SE1 and SE2. Wind power output is high nationally but lower than in other SE1–SE2 regions. The NWS target is 7.5 TWh per year, which is achievable based on the design stage. 31 "ideal sites" were identified, which had a maximum score (1) for all six social-ecological factors. These are the highest-ranked sites in terms of socio-ecological evaluation. Their sizes ranged from 2 to 10 km² with a total area of 117 km². Then, 33 leader sites were identified through non-dominated sorting. Many leaders exhibited similar factor scores, so these were grouped into four groups with four distinct factor profiles.

Four leader groups exhibited ideal factor scores (value = 1) across all socio-ecological factors, except for one factor in each group. Leader group 1 showed a slightly lower score for Wetlands (0.99); leader group 2 for Forest (0.99); leader group 3 for Cultural values (0.99); and leader group 4 for Recreation (0.92). Each leader had a set of follower sites where these followers performed inferior to the respective leader. Some leaders (or leader groups) have many followers, while others have few or none. Some followers are linked to a single leader, while others are shared across multiple leaders, as in Figure 8.Profile 1 stands out with 20 leader sites and has the largest number of followers. There were 214 exclusive followers and 349 shared followers. This means that these leaders outperform more than 90% of all sites, dominating across all factors. Other profiles perform better on other factors, where Profile 1 is weaker.Profile 1 leaders and followers show very high to ideal scores for Forest, Cultural values, and Other nature conservation values. These sites also had high factor scores for Recreation and Residential areas. There was more variation in Wetlands, although leaders still score highly.

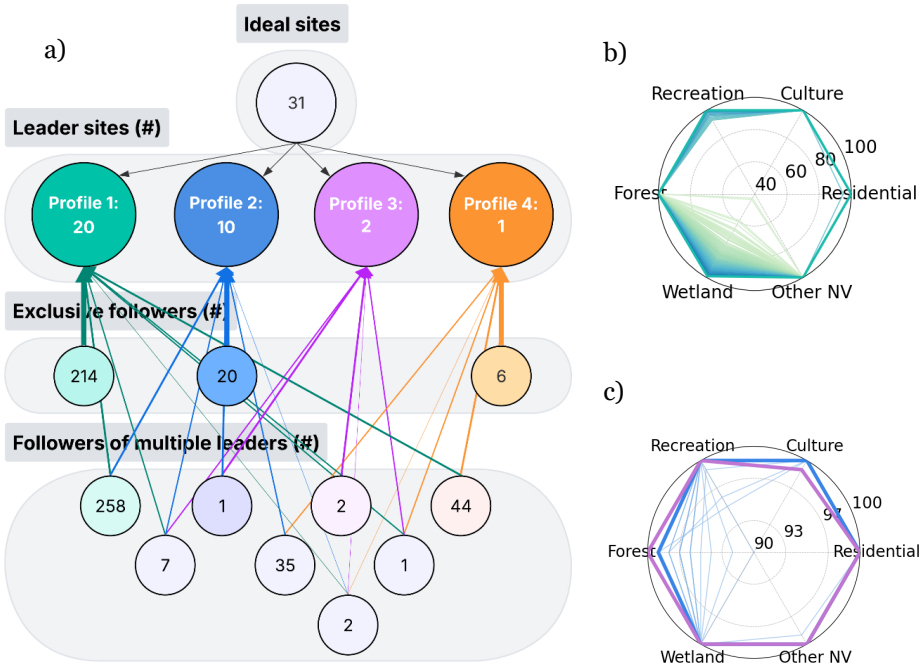


Figure 7. a) Sorting of potential sites in Gävleborg into ideals, leaders, exclusive followers and shared followers with multiple leaders (b) Profile 1 (highlighted in bold turquoise) along with its exclusive followers; and (c) Profiles 2 (bold blue) and 3 (bold purple), each shown with their respective exclusive followers.

Profile 2 consists of 20 leaders and 20 exclusive followers, and some shared followers, as in Figure 6. The exclusive followers performed very high to ideal for Wetlands, Culture and Other nature values. These sites also had high factor scores for Recreation and Residential areas. There was more variation seen in the Forest. Profile 3 consists of 2 leaders and no exclusive followers. Finally, Profile 4 consists of 6 leaders with 6 exclusive followers.

Follower sites can act as backup options if leader sites prove infeasible for other reasons. The factor scores for each region are presented separately. In Gävleborg County, sites generally have high factor scores for *Residential areas, Cultural values, and Other nature conservation values*, while lower factor scores for *Recreation, Wetlands, and Forest*.

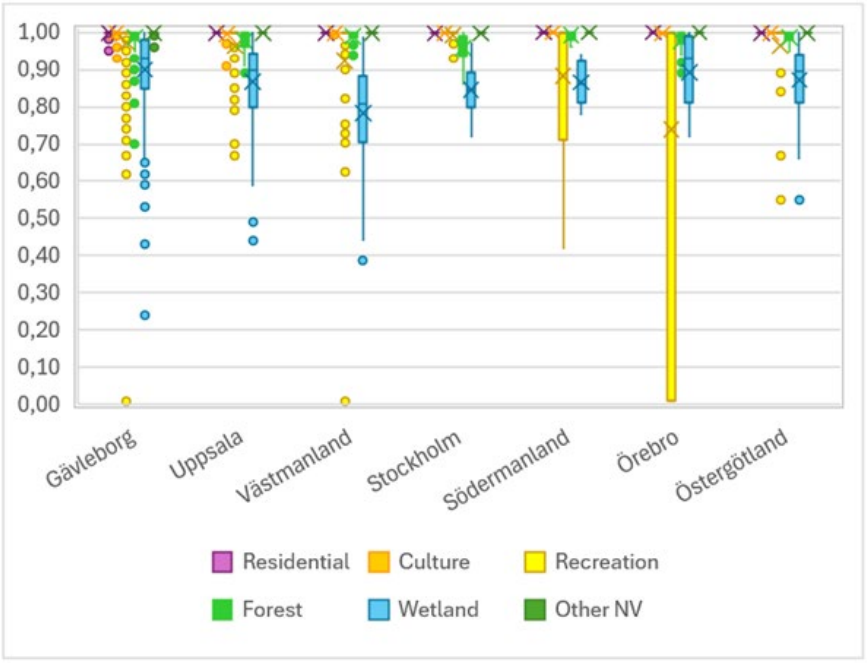


Figure 8. Overview of socio-ecological factor scores for seven regions in the study area

4.4.2 Uppsala and Västmanland

In the Uppsala Region, 82 potential sites were identified during the design stage, covering a total of 382 km² and located in the northern part of the region. Uppsala currently produces more electricity than it consumes, largely due to its nuclear power plant. However, wind power generation is limited, and many existing turbines are older with relatively low capacity. The NWS target for this region is 2.5 TWh, and the assessed potential of the identified sites at the design stage is more than three times higher than the target. In the socio-ecological evaluation, 1 ideal site (8 km²) was identified. Then 4 leading sites, each with a different profile of factor scores. Figure 9 shows very high factor scores for Residential areas and Other nature conservation values, high factor scores for Cultural values and Forest, and moderate scores in Recreation and Wetlands.

In the Västmanland Region, 76 sites were identified, covering 363 km², with individual sites ranging from 2 to 10.8 km². The sites are located in the north and north-west of the region. The region has low overall electricity production relative to demand, with minimal wind power generation. Existing wind

turbines are a mix of new high-capacity units and older installations. The NWS target is 2 TWh, and the evaluated wind power potential is approximately four times higher than the target.

No sites were classified as ideal from the socio-ecological evaluation. Two sites were identified as leaders, each showing a distinct profile of factor scores. All follower sites aligned with one of the two leaders, while the other leader had a unique profile with no followers. Overall factor performance showed clear patterns. Factor scores were very high for Residential areas, Cultural values, Other nature conservation values; high for Forest; generally medium to high for Wetlands and mostly high to medium, with a few outliers of low factor scores for Recreation.

4.4.3 Stockholm and Södermanland

In the Stockholm Region in the northern parts, 14 potential wind farm sites were identified in the design stage, covering a total of 76.7 km², with individual site sizes ranging from 2.4 to 9.6 km². The region has very high electricity demand but low production, reflecting its dense population. Existing wind power generation is low, with a mix of medium-capacity newer turbines and older, low-capacity installations. The NWS target of 2 TWh has not been achieved, although the estimated potential is relatively close at 1.7 TWh. The socio-ecological assessment showed that there were no ideal sites. Two leader sites were identified, each representing a distinct socio-ecological profile with 10 and 1 exclusive followers, respectively. Overall, factor scores were consistently high. They were very high for Residential areas, Cultural values, and Other nature conservation values; high for Recreation, Forest, and high-to-medium for Wetlands. Overall, the sites showed generally high and relatively balanced socio-ecological performance, with limited internal conflict between factors.

In the Region Södermanland, five wind farm sites were identified, covering a total of 24 km², with site sizes ranging from 2.5 to 5.8 km². The region has the lowest electricity production in the ÖMS area, supplying only about one-tenth of its consumption. Wind power generation is very limited, although two newer wind farms exist in the northern part of the region (one approved and one recently built), both with higher installed capacity. The NWS target of 2 TWh has not been met, and techno-economic results suggest that only around one-quarter of this potential could be achieved at the identified sites. No ideal sites were identified, and one leader site emerged, with the remaining four sites being followers of this leader. Notably, the leader site had a substantially higher LCoE compared to the follower sites. Factor score patterns are shown in Figure 7.

4.4.4 Örebro and Östergötland

In the Örebro Region, 31 sites were identified in the design stage, covering a total of 129 km² and ranging in size from 2 to 7.5 km². Electricity demand is more than twice the local production, and although wind power production is relatively low in an SE3 context, it remains among the higher levels within the ÖMS regions. The region includes at least nine recent high-capacity wind farms. The NWS target of 2.5 TWh has not been met; however, the techno-economic assessment indicates that the combined potential of all sites slightly exceeds this target. From the socio-ecological evaluation, one ideal site was identified, and three leader sites were identified, all sharing the same overall factor profile. Factor scores were consistently very high for Residential areas, Cultural values, and Other nature conservation values; high for Forest; high-to-medium for Wetlands; and varied from high to low for Recreation.

In the Östergötland Region, 30 sites were identified, covering 135 km² and ranging from 2.4 to 8.3 km², mainly located in the northern part of the region, with a few sites in the south near the regional border. Electricity consumption is high, approximately four times higher than production, and wind power generation is relatively low but among the higher levels within the ÖMS regions in SE3. The region includes both recent high-capacity wind farms and older, lower-capacity installations. The NWS target of 2.5 TWh has not been met, although the techno-economic results suggest that the identified sites' total potential exceeds the target to some extent. No ideal sites were found in the socio-ecological evaluation, but four leader sites were identified, forming three distinct profiles. Overall, factor scores were very high for Residential areas, Cultural values, Forest, and Other nature conservation values, while Wetlands and Recreation generally showed high-to-medium scores.

4.4.5 Techno-economic and socio-ecological evaluation

No correlation was found between socio-ecological ranking and LCoE or AEY across the sites. This means that socio-ecologically favourable sites are not by default better or worse economically, as these dimensions operate independently. Therefore, these need to be evaluated separately, and any preferences need to be guided by planning choices.

From a techno-economic perspective, it revealed clear regional differences in wind power potential relative to current production and NWS targets. Gävleborg (SE2) shows very high potential, far exceeding both present production and target levels. Uppsala and Västmanland (SE3) also have high potential, well above NWS targets, despite very low current wind power generation. Stockholm and Södermanland (SE3) exhibit low current production and low potential, with Södermanland particularly challenged in reaching

targets. Örebro and Östergötland (SE3) combine relatively high existing production with a potential that meets or slightly exceeds NWS targets.

The socio-ecological results produced a structured system of leader and follower sites, which was used to compare economic outcomes: Across most regions, there were no meaningful differences in average LCoE between leader and follower sites. Two exceptions were observed. In Gävleborg, leader sites showed slightly higher average LCoE than follower sites. In Södermanland, the single leader site had a substantially higher LCoE, though the sample size was small.

Across all regions, Recreation and Wetlands frequently received medium-to-low factor scores, even though overall site suitability exceeded the defined thresholds. This pattern likely reflects a combination of the underlying spatial distribution of landscape values and the weighting scheme used in the design stage, which shaped how these constraints influenced the final suitability outcomes.

5 Discussion

5.1 Wind power planning and its complexity in the Swedish context

Wind power development is driven by national climate and energy goals, where the benefits, such as reduced emissions and increased energy security, are largely diffuse and long-term. In contrast, the impacts are local, immediate, and long-lasting, affecting nearby communities through changes in landscape, noise, shadow flicker, property values, and land use over several decades (Sander et al., 2024, Msigwa et al., 2022). This creates a fundamental tension in onshore wind power planning, particularly regarding who bears the burdens and who receives the benefits. It is often less populated and less developed areas in northern Sweden that host a larger farm with taller turbines, while electricity through the national power grids is transmitted to and consumed in southern Sweden (Vindbrukskollen, 2026). This raises concerns regarding distributional justice, as the benefits and burdens of wind power development are not equally shared (Liljenfeldt and Pettersson, 2017, Lindvall, 2023). At the same time, procedural justice concerns arise when local communities feel they have limited influence over decisions that significantly affect their surroundings, which can contribute to strong local opposition (Tsani et al., 2024).

Beyond social acceptance, institutional and technical constraints further complicate the wind power planning in Sweden. The Swedish Armed Forces may also restrict developments due to defence-related interests, and these restrictions cannot be appealed against (Odell et al., 2022). In addition, grid capacity constraints and long approval timelines further increase uncertainty for developers. As a result, there is often a gap between strategic planning at broader scales and local decision-making, where political, social, and institutional factors ultimately determine feasibility.

5.2 Role of SMCA in wind energy planning

SMCA provides a useful framework for structuring complex wind farm siting problems by integrating multiple spatial criteria into a single analytical process. It helps organise large amounts of spatial information alongside stakeholder input, producing composite indicators that can support decision-making. These criteria often represent diverse disciplines, and the stakeholders affected may come from different interest groups. Through SMCA, these diverse aspects can be brought together in a systematic way. In this sense, SMCA simplifies complex decision contexts and improves transparency by structuring both data and stakeholder inputs.

However, wind energy planning involves inherently conflicting site-specific criteria and rapidly evolving concepts of sustainability that can vary significantly across stakeholders, locations, and time periods. In addition, differences in methodological steps and choices across studies lead to variation in overall approaches and outcomes.

SMCA is applicable across multiple stages of planning. At broader scales, it helps identify suitable areas across large regions, while at more detailed levels, it enables the comparison and evaluation of specific alternatives.

Overall, the findings indicate that while SMCA provides valuable insights, there remains considerable scope to improve its applicability in real-world planning contexts, particularly in addressing the complexities of stakeholder involvement and decision-making under uncertainty.

5.3 Limitations of SMCA in practice

The results from this thesis (particularly Papers 2 and 3), together with the review presented in Paper 1, indicate several limitations in the application of SMCA in wind energy planning. The combination of AHP and WLC is the most used approach and is often highlighted for its ability to structure complex problems into hierarchical frameworks, facilitate intuitive pairwise comparisons, and include consistency checks to assess the reliability of judgments. Apart from simplicity in the process and assumptions, the availability of WLC in several GIS software platforms, such as TerrSet and ArcGIS, also makes the method easy to implement in the GIS domain.

5.3.1 Weighting exercise and complications

There are different weighting approaches, each with inherent trade-offs between the cognitive burden of numerous comparisons and the simplicity and ease of use of more straightforward approaches. There is no single weighting method that can be considered universally best (Marttunen et al., 2015).

The practical application of the AHP and RR methods in focus group settings revealed several challenges. The number of required pairwise comparisons in AHP became excessive, leading participants to shift their focus from sustainable wind power planning objectives to maintaining acceptable consistency ratios. In addition, the hierarchical structuring of criteria introduced splitting bias (Montibeller and Winterfeldt, 2015), whereby the organisation of the planning tree itself influences the resulting weights rather than purely reflecting stakeholder preferences. These challenges have also been discussed previously in other literature (Asadabadi et al., 2019, Rezaei, 2015, Marttunen et al., 2018).

Furthermore, the subjectivity of weights derived from different stakeholders remains a persistent challenge of SMCA, as results can vary significantly depending on who is involved in the weighting process (Malczewski and Rinner, 2015). Beyond the weighting procedure, several broader limitations of SMCA were identified across the papers.

5.3.2 Spatial indicator limitations

A recurring issue is the reliance on "borrowed" simplified spatial indicators that act as proxies for complex social, ecological, and technical realities. Furthermore, social acceptance and community attitudes towards wind farms, arguably among the most critical factors in wind energy planning, are difficult to capture spatially and are rarely represented adequately in SMCA frameworks (Tsani et al., 2024). This is reflected in the Paper 1 review, which found that publications focus on technical feasibility and infrastructure considerations. However, studies from Sweden show that wind farm approvals are to a large extent shaped by socio-economic and political factors beyond technical suitability (Liljenfeldt and Pettersson, 2017, Rahmad et al., 2025).

As discussed in Papers 1 and 2, other important factors such as available grid capacity, ecological sensitivity, and wind resource are either oversimplified or represented using limited datasets, often due to data availability constraints. We encountered similar challenges in this thesis when collecting, interpreting, and standardising input data across Papers 2, 3, and 4.

First, regarding techno-economic factors, we faced several data-related difficulties. Power grid data were incomplete, particularly regarding reserve capacity, substation locations, and planned grid developments, which limited the ability to realistically model future connection possibilities. For wind resource assessment, questions arose about which data and hub height to use, especially given the rapid development of increasingly tall turbines.

Second, there are uncertainties in how social dimensions should be spatially represented and interpreted. Criteria such as residential exposure, visual impact, and proximity to human activities raise methodological questions about whether they should be treated as strict constraints or as continuous suitability

gradients, and whether all sensitive land uses (e.g., housing, hospitals, holiday homes) should be weighted equally or differentiated. In addition, questions remain regarding the accuracy and relevance of detailed viewshed and noise propagation modelling at the planning stage, particularly when turbines can, in theory, be placed anywhere within the study area.

Another was the reindeer herding, representing a critical land-use factor, yet access to up-to-date, spatially detailed data remains limited despite established research in this area. In addition, defence-related spatial data are often incomplete or restricted due to confidentiality, further constraining comprehensive assessments.

5.3.3 Implications of standardisation

Standardisation of factors into a common scale is carried out by the modeller based on inputs from literature and/or stakeholders. These inputs typically relate to thresholds, such as safe/buffer distances (e.g., from roads or power grids), minimum requirements (e.g., minimum wind speed, maximum feasible distance to power grids), or limits beyond which impacts are assumed to be negligible (e.g., distance from houses). In addition, the modeller must define value or fuzzy functions that specify the functional relationship between criterion values and suitability, including how suitability changes across ranges in linear, exponential, or other forms. Linear standardisation is commonly applied, which may not adequately represent non-linear real-world relationships (e.g., wind speed and energy production). What makes this process particularly important is that it involves translating highly diverse and incomparable factors into a single suitability scale. While this scale can be conceptually defined in verbal terms, its formal unification into a numerical representation is inherently challenging. This step, therefore, represents a critical point at which values, judgments and biases are first embedded into the model.

For these reasons, it is important to consider whether these decisions are grounded in scientific evidence or informed by local stakeholders' experiences. Residents, particularly those with prior experience of living near wind turbines, may perceive impacts differently from those suggested by technical assessments. As a result, there is often no single correct standardisation approach. Instead, the translation of these inputs into factor maps can follow a more precautionary (defensive) approach, which sacrifices larger areas to minimise potential impacts, or a more permissive (offensive) approach, which allows greater flexibility but may accept higher levels of risk.

Given this subjectiveness, transparency in the standardisation process is essential. The assumptions underlying thresholds and value functions should be clearly documented, allowing decisions to be traced and evaluated. It should be

possible to identify whose knowledge and perspectives have influenced these choices and how they are weighted within the model. Such transparency is crucial for ensuring the credibility and legitimacy of the results.

5.3.4 Trade-offs and interpretability

In Papers 2 and 3, applying WLC to aggregate factor maps and stakeholder preferences into a single suitability score leads to a significant loss of information and reduced transparency. It becomes difficult to trace how trade-offs between factors are resolved, especially when a final suitability threshold is applied. What ultimately drives the outcome and the interaction between factor scores and stakeholder preferences remains largely opaque.

Moreover, expressing stakeholder preferences as numerical weights is itself conceptually challenging. When these weights are applied to standardised factor maps and aggregated, additional layers of abstraction are introduced. While the process may appear mathematically straightforward, it is conceptually difficult to interpret. The combined effects of standardisation, weighting, and additive aggregation make it hard to understand how individual factors contribute to the final suitability score and the extent of compensation or trade-off between factors that occurs (Malczewski, 2000, Becker et al., 2017).

In addition, when factors are interdependent, which is often the case in complex spatial planning problems, there is a risk of double-counting. WLC theory typically recommends eliminating such interdependent factors (Malczewski, 2000). From a planning perspective, however, stakeholders often consider all factors relevant, even when they are interrelated. In the real world, the planning goal is not only to produce an optimal mathematical solution in SMCA but also to represent multiple coexisting concerns.Furthermore, the ranking of best alternatives based on a single aggregated score may introduce omission bias (Montibeller and Winterfeldt, 2015). When only the top-ranking alternatives are selected and discussed further, options that perform well from a specific planning perspective may be overlooked.

Taken together, these issues suggest that weights and additive aggregation can introduce further conceptual and practical challenges. The results from Paper 4 further demonstrated the importance of delaying the aggregation of techno-economic and socio-ecological perspectives, as these dimensions showed no correlation across the evaluated sites. Therefore, combining these dimensions into a single suitability score and applying weights could hide trade-offs and reduce the interpretability of planning outcomes.

5.3.5 Limited sensitivity analysis

To make these invisible trade-offs more transparent and to clarify the role of weights in the decision-making process, a systematic sensitivity analysis (SA) is

recommended within SMCA. Such analyses help identify which factors are most influential in each case, thereby supporting participants in building confidence in the robustness of the results. In addition, visualising weight sensitivity geographically has been highlighted as particularly valuable (Feick and Hall, 2004).

However, findings from Paper 1 show that sensitivity analysis was conducted in only 46% of the reviewed studies, and among these, nearly 68% relied on simple weight alterations. This suggests that, in most cases, sensitivity analysis is applied in a rather limited manner and does not fully explore the implications of uncertainty in weights.

At the same time, more advanced approaches, such as global sensitivity analysis or detailed one-at-a-time sensitivity analysis, are not seen to be applied in the reviewed literature. Similarly, sensitivity analyses addressing criteria formulation or the use of alternative SMCA methods are only considered in a small number of studies. While these approaches can provide deeper insights, they also introduce additional complexity, generating multiple scenarios and layers of information that may be more difficult to interpret than the original factor score distributions and stakeholder preferences.

5.3.6 Fragmented stakeholder involvement

Turning to stakeholder involvement, the findings from Paper 1 indicate that stakeholders are only involved in selected stages of the SMCA process. Key steps, such as defining the decision hierarchy or determining the suitability threshold (as in the design stage), are generally carried out without their participation. There is some, but very limited, involvement in the standardisation of criteria. Standardisation of factors and participants' perceptions of the suitability scale can themselves be biased. In addition, stakeholder judgements may be influenced by their professional expertise and prior experiences, leading to greater emphasis on criteria that are related to their own areas of knowledge (Montibeller and Winterfeldt, 2015).

For SMCA to effectively support participatory planning, stakeholder involvement should extend beyond weighting to encompass the different stages of the decision-making process, such as defining the decision problem, criteria selection, weighting, aggregation, uncertainty analysis and interpretation of results (Thokala and Madhavan, 2019). When assigning weights, participants should be aware of how factor scores are distributed and how their weighting decisions influence the overall suitability outcome. However, such integrated involvement across the SMCA workflow is rarely reflected in the reviewed Wind-SMCA studies.

5.4 Methodological contributions

5.4.1 Conflict score (Paper 2)

The conflict score was developed in Paper 2 to address the limitation of the invisibility of trade-offs in WLC outputs. In WLC, high weights can disproportionately influence the final suitability score and compensate for low values in individual factor maps, reducing transparency in how factors interact in the aggregated result.

To make these trade-offs more explicit, a conflict-based measure was introduced using an inverse formulation based on deviations from maximum values, conceptually like a standardised deviation measure. Through transformations such as inversion or squaring, low factor scores are emphasised, making areas of weak performance more clearly visible rather than being masked by weighted aggregation. This improves interpretability compared to WLC.

Suitability and conflict represent two complementary dimensions of spatial decision-making. Suitability describes the overall desirability of a location, while conflict reflects the degree to which favourable and unfavourable conditions coexist. A highly suitable area is therefore not necessarily a low-conflict area, as suitability may arise through compensatory effects between opposing factors.

5.4.2 RR method (Paper 3)

The RR method applied in Paper 3 was perceived by participants as straightforward, transparent, and easy to use in practice. Its two-step structure, first ranking factors and then refining preferences through pairwise comparisons, was considered intuitive and supportive of structured reflection on preferences. The stakeholders found it easier to rank factors than to judge the exact magnitude of differences among preferences between factors, even with the help of the Saaty scale. They tend to focus on the ranking step and spend more time on discussions there, while pairwise comparisons went faster.

Compared to more complex approaches such as AHP, the method reduced procedural complexity while still enabling a systematic expression of stakeholder priorities. It also helped maintain focus on factors and their relative importance throughout the discussion.

The cluster-based overview was viewed as particularly useful for maintaining an overall understanding of the planning structure during the weighting exercise. It facilitated discussions by providing a clear picture of how impacts were distributed across different clusters. This is an inherent step in the RR method, and it helps stakeholders reflect on the final weights based on the planning tree structure. Through this, stakeholders are encouraged to compare factors based on different clusters rather than only focusing on comparing with the least

important factor, thereby reducing the anchoring bias (Rezaei et al., 2024, Montibeller and Winterfeldt, 2015).

In addition, the factor maps, which stakeholders were already familiar with prior to the exercise, were considered helpful for understanding the spatial implications of the resulting weights and their overall effects.

Overall, the RR method contributed to improved participant engagement and a stronger sense of control over the weighting process. Its iterative nature supported dialogue and reasoning about preferences in ways that align well with participatory planning contexts. Two potential improvements were suggested: (i) expanding the number of available scales, or (ii) allowing participants to first group factors into broader importance categories (e.g. high, medium, low) and then conduct weighting within each group in a second step.

5.4.3 The leader-follower approach (Paper 4)

The application of the leader–follower approach in the socio-ecological evaluation offers a more transparent way of analysing alternatives without aggregating spatial data or weights into a single score. Using non-dominated sorting, sites were compared only on their factor scores. Standardisation was, however, carried out to simplify the process and ensure comparability of indicators rather than relying on raw criterion values. Rather than producing ranks or a composite indicator, this approach encourages discussion on the overall distribution of factor scores across the study area and how these should be interpreted at a larger spatial scale.

In participatory settings, presenting only the top-ranked sites with suitability scores can steer discussions towards individual locations and their specific problems. This may limit the exploration of broader planning strategies and reduce attention to other alternative or backup options. In contrast, the leader–follower structure allows for a wider perspective by keeping multiple viable alternatives visible. Some regions contained ideal sites, while others had only leader sites, with or without follower sites. Setting aside ideal sites, leaders represent the best available options and comparing them is a first step in understanding the overall opportunities within the study area.

Followers, which are weaker than leaders, are visualised using radar diagrams, making all alternatives and their factor scores explicit. This representation not only highlights variation across factors but also illustrates where trade-offs become apparent, as indicated by intersecting patterns in the diagrams. Once leader and follower options are exhausted, additional Pareto fronts can be identified, revealing further layers of alternatives. Overall, the leader–follower structure improves interpretability by translating multidimensional results into a clear planning language of ideal, leader, and follower sites, thereby supporting more transparent and flexible decision-making in wind power planning.

6 Conclusions

Onshore wind power planning in Sweden is characterised by high complexity, long permitting times, multiple governance levels, power grid constraints, distributional justice concerns and strong institutional influence from both municipalities and defence interests. These conditions contribute to frequent project uncertainty and rejections, highlighting the need for robust planning support tools capable of handling conflicting spatial factors and constraints, as well as facilitating a platform where regional, municipal authorities and local perspectives can meet and discuss to reach a negotiated decision.

In this regard, SMCA provides a relevant decision-support framework. This thesis set out to improve the applicability of SMCA for onshore wind energy siting in Sweden. The results show that SMCA outcomes are strongly shaped by methodological choices, particularly in the selection of criteria, standardisation, weighting, and aggregation. These steps inevitably introduce subjectivity, which becomes problematic when analytical processes lack transparency and when results are reduced to map outputs showing suitability scores or ranks of suggested sites, which can be seen as over-simplified representations of site suitability. This reduces interpretability and makes it difficult to trace how different values and stakeholder preferences influence final outcomes.

In onshore wind power planning, where decisions are socially consequential to a larger area and broader society, this limitation can weaken trust and reduce practical usability. As echoed throughout this thesis, the key requirement for effective decision support is not to increase methodological complexity but to ensure simplicity, transparency, and interpretability in the analytical procedure.

The thesis also highlights that SMCA applications are highly dependent on the availability and quality of input data, as well as cross-sectoral collaborations. In practice, we encountered limitations in key datasets, including power grids, substation infrastructure and capacity, reindeer herding areas, and sensitive bird species, all of which are influenced by spatial and temporal variability. These challenges underscore that SMCA depends on cross-disciplinary collaboration, and that insufficient integration of interdisciplinary knowledge risks producing incomplete or biased representations of planning reality.

To address these challenges, the thesis develops several methodological contributions. A conflict score was introduced to make trade-offs more explicit. The RR method improved transparency and usability in participatory weighting processes. The leader-follower approach is an attempt to move away from single suitability scores and rankings of alternatives, toward structuring planning alternatives so stakeholders can decide based on factor profiles. In addition, separating techno-economic and socio-ecological evaluations strengthened interpretability by clarifying that these dimensions are better considered separately, leading to more meaningful discussions and understanding.

7 Future research

Future research could focus on simplifying and structuring diverse spatial information in a meaningful way, enabling stakeholders to first obtain an overview of the situation, then identify patterns, understand conflicts, and ultimately make informed decisions that can be traced back to the underlying information. Although several papers have been published on the Wind-SMCA topic, relatively few studies explicitly discuss the trade-offs that occur within the study area. The leader-follower approach developed and applied in this thesis is an attempt in this direction and requires further application across different study areas, spatial scales, and stakeholder groups. Future research should also explore how stakeholder interests can be integrated into such approaches, potentially not only as numerical weights, so that the information revealed by the approach is not constrained or obscured by the weighting process. In this context, further investigation of the RR method is needed, including exploring how factors can be grouped by importance and how such grouping influences stakeholder preferences. In this study, all factors were standardised to a common scale during the evaluation; however, preserving original values could provide a more direct representation of the underlying information. Furthermore, while this thesis emphasises transparency and clarity in decision-support processes, some datasets related to defence interests and critical infrastructure require controlled disclosure. Future research should therefore investigate how such approaches can provide meaningful insights and support decision-making while protecting sensitive information.

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